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ICT: CONCEPTUAL FRAMEWORK

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UNIT 5 ICTS FOR DEVELOPMENT: AN OVERVIEW

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5.0 INTRODUCTION

It is an established fact that use of Information and Communication Technology (ICT) in development by the developed nations is more as compared to developing nations. In this unit, you will read about the role of ICT in development.

5.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- Define the meaning and explain attributes of ICT;
- Explain ICT and development interface;
- Narrate the role of ICT in development of various sectors; and
- Comprehend e-Development strategies.

5.2 ICT: MEANING AND ATTRIBUTES

5.2.1 Meaning of ICT

Customarily, ICT refers to *information, communication and technology*. Generally we seek information and communicate them to others who may also be interested to know about its value through technology. The *technology*

plays an important role in the effective transaction and communication of knowledge or information to its seekers. ICTs include hardware, processes and systems that are used for storing and managing communications and sharing of information. These tools can be either manual or computerized. ICT is defined by the United Nations Development Programme (UNDP) as, “ICTs are basically information handling tools, a varied set of goods, applications and services that are used to produce, store, process, distribute and exchange information. They include the ‘old’ ICTs of radio, television and telephone, and the ‘new’ ICTs of computers, satellite and wireless technology and the Internet. These different tools are now able to work together, and combine to form our ‘networked world’ – a massive infrastructure of interconnected telephone services, standardized computing hardware, the Internet, radio and television, which reaches into every corner of the globe”. The ICT has enabled ICT using nations to become an *Information Economy*. The UNDP document defines *Information Economy* as “a new global electronic structure where in the production of information goods and services dominates wealth and job creation and is underpinned by the use of information and communication technologies (ICT) and the global infrastructure”. The OECD member countries define the ICT sector as a combination of manufacturing and services industries that capture, transmit and display data and information electronically. Some commonly used ICT tools are described:

- i) **Mobile phones:** An electronic, portable and wireless communication device that is available in different shapes, sizes and models. Mobile phones connect to a wireless communication network through radio waves or satellite transmissions. According to business dictionary “Mobile Phone is a portable device that does not require the use of landlines. It utilizes frequencies transmitted by cellular tower to connect the calls between two devices”. Mobile phones are capable of communicating via voice calls, e-mails, text messages, video, picture messages, blue tooth, infra-red and fax.
- ii) **Video Conferencing:** According to the BSNL, video conferencing services allow multiple participants to converse with each other regardless of their location through the video end-points. It enables new ways for communication and collaboration between people at almost any location within the world. It involves the transmission of visuals and sounds to two or more separate locations through the use of cameras, monitors, speakers and microphones. Multiple video-conferencing enables three or more individuals to sit in a virtual conference room and communicate as if they were sitting right next to each other.
- iii) **Radio:** The radio is the wireless transmission through space of electromagnetic waves in the approximate frequency range from 10 kilohertz to 300,000 megahertz. Radio is a user friendly and cost effective technology. It is an effective communication and knowledge sharing tool.
- iv) **Internet:** Internet is a global network and is a revolution in IT which is connecting millions of computers. The Internet has become a public, cooperative and self-sustaining facility available and accessible to

millions of people, Non Governmental Organizations (NGOs), offices, civil society organizations, private corporate sectors, etc., Worldwide. The Internet carries a vast array of information resources and services, most notably the inter-linked hypertext documents of the World Wide Web (WWW) and the infrastructure to support electronic mail (e-mail), in addition to popular services such as video on demand, online shopping, online gaming, exchange of information from one-to-many or many-to-many by online chat, online social networking, online publishing, file transfer, file sharing and Voice over Internet Protocol (VoIP) or teleconferencing, tele-presence, person-to-person communication via voice and video.

- v) **Television:** It is a widely used telecommunication medium for transmitting and receiving static and moving images usually accompanied by sound. Televisions are available in different sizes and with many advance features, including Internet facilities.

The ICT has established that there is an inherent relationship between information, communication and technology. The synergistic relationship between information, communication and technology is described in Fig. 5.1.

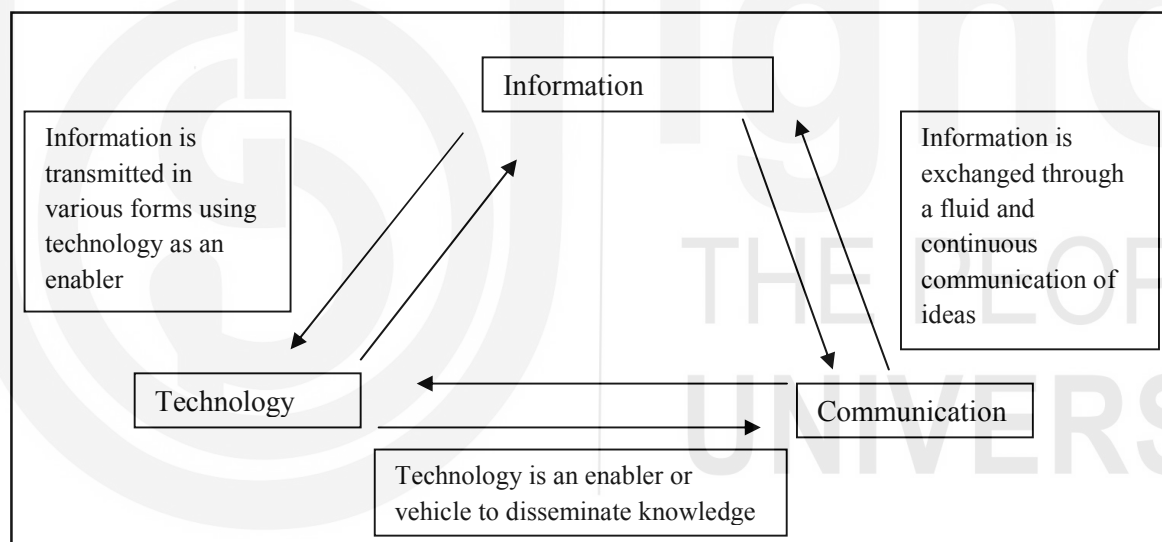


Fig. 5.1: Synergistic Relationship between Information, Communication and Technology

5.2.2 Attributes of ICT

After knowing the meaning of ICT, it is essential to know about its attributes. Some of the important attributes of ICT are:

- i) **Knowledge Revolution:** ICT has dramatic power to organise information. This has also accelerated the pace of learning, innovation and knowledge creation and dissemination. It has helped knowledge power in a great way. The transfer of knowledge has been faster after the advent of ICT. ICT and the growth of knowledge society have become almost synonymous.
- ii) **Productivity Revolution:** The ICT has impacted the ICT using industries and services in raising the overall productivity. The economically

advanced country USA, which has effectively used ICT for enhancing the productivity across various sectors of the economy, is one of the illustrious examples. In India also, sectors that are using ICT effectively are able to raise productivity much faster than their counterparts. ICT in most of the service sectors has reduced cost and enhanced productivity and output.

- iii) **Learning Revolution:** According to World Bank, ICT has created a learning revolution that has given rise to lifelong learning. According to Resnick, ICT empowers the students becoming more active and independent learners. ICT has made the people to be less dependent on class room teaching.
- iv) **Promoting Connections:** ICT has promoted connections among people, NGOs, enterprises and communities. ICT gives rise to empowerment, participation, coordination, decentralisation, social learning, connecting communities of practice, mobilising social capital and globalising civil society concerns. ICTs have been increasingly described as *technologies of freedom*. ICT is empowering people to connect, mobilise, organise, overcome their isolation and share their experiences and idiosyncratic information.
- v) **Innovation-Driven Economy:** The ICT is an enabler to transform the nation from investment-driven economy to innovation-driven economy which is more efficient and productive. ICT has fastened the innovation process in scientific societies.
- vi) **Globalisation booster:** ICTs have been a key engine for the performance and growth of economies since the early 1970s, becoming of course the main technological enablers of economic globalisation. The global information has become easily accessible through ICT. It is one of the important energizers of globalisation and has made the world a global village.
- vii) **Promoter of Human Development:** The United Nations has long recognised the need for timely and relevant information as a fundamental element of human development. It has emphasised on universal access to information and communication services as a basic need for human development. The human development is effected by digital divide.
- viii) **Contributer to Research and Development:** ICT has contributed greatly to research and development. Research guides and scholars are using ICTs as tools for acquiring information and knowledge. The research and development activities of various sectors and institutions have been benefited immensely by the ICT revolution.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) What do you mean by ICT?

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2) Write short notes on Internet as an ICT?

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5.3 ICT AND DEVELOPMENT INTERFACE

Since the early 1970s the pace of ICT has been recognised as a key engine for enhancing development and accelerating growth of economies. ICTs have the potential to not only improve the delivery system of government services but also enhance development outcomes through greater transparency, ease of access, people's participation and cost reduction. While encouraging the role of ICT in development Kofi Anan, the former Director General of UNO said, "A technological revolution is transforming society in a profound way. If harnessed and directed properly, ICTs have the potential to improve all aspects of our social, economic and cultural life. ICTs can serve as an engine for development in the 21st century". The UN has recognised the need for timely and relevant information as a fundamental element of human development and has called for universal access to information and communication services as a basic development need. While delineating the role of ICT, the Asian Development Bank (ADB) has emphasized that "ICT for development is not about computers, mobile phones and the internet, but about help, support and training people in linking them and communities for communication, learning and services. This will lead to improved well-being, increased work productivity, support for innovation and impetus for inclusive growth". The thrust areas of ICT strategies for development as envisaged by ADB is to create an enabling environment by fostering the development of innovative sector policies, strengthening public institutions and developing ICT facilities and related infrastructure and network. World Bank states "Information and communication technologies are a key input for economic development and growth. They offer opportunities for global integration while retaining the identity of traditional societies. ICT can increase the economic and social well-being of poor people and empower individuals and communities. Finally, ICT can enhance the effectiveness, efficiency and transparency of the public sector including the delivery of social services". The ICT for development need to emphasise the following dimensions of ICT:

- i) **Information-centric:** Focusing more on information and data and less on technology. In the knowledge society, information and knowledge

is considered to be more powerful and is an important tool for empowerment. The Right To Information (RTI) Act in India empowers common citizens to get official information and play a pro-active role in governance.

- ii) **Chain-centric:** The ICTs need to provide the connected data relating to the main problem. For example, if health status of a village is low, the ICT need to provide data on infant mortality rate (IMR), institutional delivery, immunisation status, disease prevalence, etc., which are the determinants of health status. Sporadic information is dangerous and will cause more harm than benefit. It will not help the decision maker to take appropriate decisions.
- iii) **Society-centric:** The ICTs need to provide data on social aspects such as health, education, status of women, role of youth in development, etc. Partial information is always dangerous and cannot be helpful to take corrective measures.
- iv) **Economy-centric:** The ICTs not only provide data and information about economic development but also enable new or more productive income generating activities. For example in Bangladesh, through the national phone operators, villagers purchased their phones as members of the Grameen Bank, get accessible pay-phone services. The information about various development schemes, welfare programmes, and successful projects can be communicated to the rural people through ICT.
- v) **Development-centric:** ICT in order to promote development should be development centric. The ADB is helping its member countries in integrating ICT components in sector development strategies especially in education, health and agriculture. The focus is on improving public administration and finance management, as well as providing various electronic services to citizens and businesses. The ICT should provide information about performance of various sectors of development and different programmes of development.
- vi) **Manpower-Centric:** ICT plays an important role in manpower development. A study claims that ICT diffusion accounts for up to 90 percent increase in Human Development Index of some of the nations. Therefore, most of the ICT proponents suggest that ICT should focus more on people rather than just human efficiency and ultimately raise their knowledge and productivity. For this, at first the requirement of the people who are to be benefited from ICT needs to be understood and the ways in which ICT can be helpful must be chalked out.

The ICT enabled “Information Chain” for development is given in Fig.5.2

Data Resources	Information	Decision	Action	Impact	Result
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Fig. 5.2: ICT Information Chain

The three types of ICT enabled development are:

- i) Direct Development
- ii) Networked Development
- iii) Grassroots Development

i) **Direct Development:** In this model, the ICTs deliver resources and services directly to the beneficiaries without any interventions of other development actors like NGOs, PRIs and SHGs, etc. The direct telecast of many development programmes of the government through ICT is a direct development endeavour. This is a one-to-one channel as depicted in Fig. 5.3.

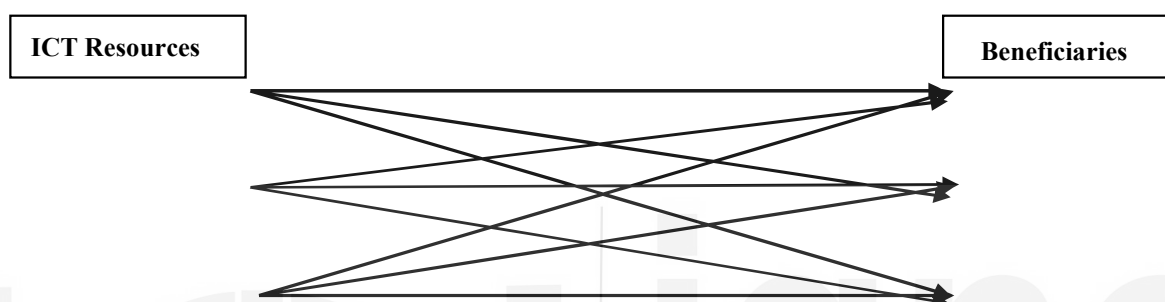


Fig. 5.3: ICT and Direct Development

ii) **Networked Development:** In this model, instead of directly providing information or data to the beneficiaries, the state or private agencies act through other actors or institutions that are connected and can effectively act through ICTs. Here the main source is inter-connected. These intermediaries are not only inter-connected with the main sources but also inter-connected among themselves. The example is given in Fig. 5.4

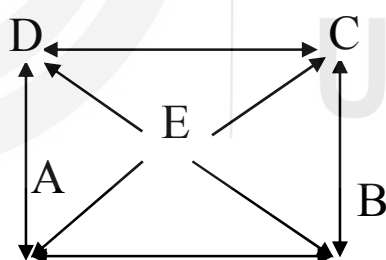


Fig. 5.4: ICT and Networked Development

iii) **Grassroots Development:** In the grassroots development model, the main development agency, for example, government takes the help of NGOs and other civil society organisations to reach to the beneficiaries. Now a days, the state as an actor of development gradually receding its onus to the community level organisation or civil society organisations. There are several successful examples of community development models. It is a democratic way of pursuing development. The grassroots development model is given in Fig. 5.5

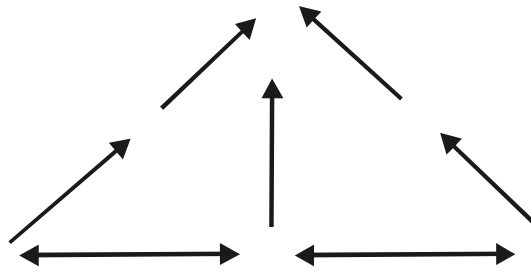


Fig. 5.5: ICT and Grassroots Development

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) ICT should be Development centric-Justify?

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2) Write short note on ICT enabled grass roots model?

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5.4 ICT AND SECTORAL DEVELOPMENT

ICTs play an important role in enabling the nation to achieve the objective of Millennium Development Goals. The economic, social and human development needs of the developing nations in particular cannot be met in timely and effective manner without the innovative and strategic use of new technologies. The *digital divide* is also one of the reasons for differences in the percentage of achievements in various development indicators in both rural and urban areas. It is remarked that the *digital divide* is really a reflection of existing disparities between the haves and have nots. According to the Organisation for Economic Cooperation and Development (OECD), “The digital divide is a symptom of existing economic and social divide, which will widen even in future, if developing countries are not helped to take advantage of ICT in tackling economic and social problems and are denied access to markets that are becoming increasingly ICT-dependent as a

part of globalisation”. Internet access is one of the criteria in which we can assess the reach of ICT in the global context Figure 5.1 describes the status of ICT reach.

The discrepancy in urban and rural development is a cause of concern for many developing countries. One of such discrepancies is higher literacy rates in urban areas as compared to rural areas. E-Development, therefore should aim at harnessing the ongoing technological revolution to achieve Sustainable Development Goals, one of which is certainly of raising literacy and educational status. E-Development would systematically address the opportunities to use ICT for the competitiveness of developing economies and to expand employment and earning opportunities, to access market information and lower transaction costs for the poor, women, marginalized communities, farmers, traders and artisans. It will be used to achieve higher agricultural and industrial growth rate and enhance literacy and health status of the country’s population. However, it is observed that many of the developing countries are still lagging in the use of ICTs in various sectors of development. One of the findings on the ICT role in development reveals that the technological information infrastructure has failed to link the growing awareness of the importance of knowledge for development to the key actors and the information seekers who can carry forward the development actions bridging ICT. The optimal and innovative use of ICT tools will have a decisive bearing on the success or failure of development. This decisive role that ICT plays in facilitating successful implementation of development objective is clearly spelt out in the declaration of principles of the World Summit on Information Society. The role of ICT in development of various sectors is narrated as under:

5.4.1 ICT and Poverty Reduction

ICTs play an important role in poverty reduction strategies adopted by various countries. One of the main opportunities for using ICT in poverty reduction is to provide information and knowledge to rural populations and to empower local development agents to serve the poor. Lack of knowledge regarding various poverty alleviation schemes and their procedures of implementation are important reasons for poor implementation of governmental programmes particularly, the rural development programmes. The illiterate people have poor access to information and therefore, are unable to tap the benefits meant for their upliftment. To sight a few examples, the Chile’s electronic rural information system has enabled development actors such as NGOs, municipalities and extension agencies in transmitting information on prices, markets, inputs, weather, social services, and credit facilities at a cost 40 percent less than that using traditional methods. Another example, the land record computerisation in Karnataka, India has enabled the agencies to deliver land certificates within 15 minutes instead of 20–30 days and in the process, has reduced transaction costs and corruption, created a viable land market and enhanced the quality of life of the common man.

The tele-centre or the community information and communication centres play important role in transacting information about various development initiative programmes of the government to the common men. These centres also have enabled rural community to carryout local dialogue, share practical locally relevant information and support community problem solving. According to M. Fontaine, digital literacy centres in Benin and Ghana have become an important instrument of empowerment of low-income communities, enhancing employability, increasing capabilities and extending learning opportunities beyond those available in education institutions. While justifying the role of ICTs, Mohammed Yunus, the architect of microfinance in Bangladesh remarked that “there is an on-going view that IT is totally irrelevant for the poor who are generally illiterate; IT is too expensive for them to reach out; the poor do not need fancy IT, they need food. These are the voices of the sceptics. Now in three years, there are more than 5000 telephone ladies in Bangladesh villages doing roaring business in selling telephone services.” Some of the lessons learned from the use of ICTs for poverty reduction are:

- i) Technologies used must be adequate to the skills of the poor in order to exploit their potential effectively, in other words it must be user friendly and pro-poor;
- ii) Content should receive as much attention as connectivity and it must be people-centred, demand-driven and in local languages; and
- iii) Ownership by the local communities, partnership and networking is key to effective poverty reduction programme. The local NGOs and civil society organisations are required to be adequately involved through ICT for the poverty reduction at the grassroots level.

5.4.2 ICT and Agriculture

Agriculture is the primary sector of development. ICT in agriculture is an emerging field focusing on the enhancement of agricultural productivity and growth rate. The technological revolution could benefit the agriculture sector through increased flow of information, transfer of capital and inputs and quality of various services provided to agriculture sector. The role of ICT to enhance food security and support rural livelihoods is increasingly recognized and was officially endorsed at the World Summit on the Information services (WSIS, 2003–05). The ICT can potentially help the farming community in the following ways:

- i) Meet the information, technical and input needs of the farmer more efficiently;
- ii) Improve access of knowledge and technology among farmers and farmers linked institutions;
- iii) Strengthen dialogue between and among farmers and farm-linked institutions;
- iv) Promote interactive learning among the farmers and other local based

organizations such as *panchayats*, milk cooperative societies WSHGs and village farmers groups.

All these will lead to increase in farm outputs or greater value addition of farm products, resulting in increase in household incomes and quality of life of the farmers. Further, ICTs can promote trade and competitiveness of agricultural products and broaden markets for agro-products leading to increase in domestic GDP. The FAO's *e-Agriculture* initiative aims at ensuring systematic dissemination of information using ICTs on agriculture, animal husbandry, fisheries, forestry and food, in order to provide ready access to comprehensive up-to-date and detailed knowledge and information, particularly in rural areas.

The ICTs nowadays play an important role in the promotion of public private partnership (PPP) in the area of *ICTs for Agriculture Development*. The four types of partnership which are desirable for agricultural development are:

- i) Providing affordable ICT access and connectivity to farmers and farm-linked institutions;
- ii) Providing relevant contents to the experts, trainees and extension agents;
- iii) Training and capacity building;
- iv) Projecting the locally available good practices;
- v) Promotion of contract farming through public-private partnership.

There is a need to understand as to how far the ICT initiatives are able to address the farmers' needs as far as agriculture development is concerned.

5.4.3 ICT and Education

ICT is a powerful tool for extending educational opportunities both formal and non-formal to all sections of the society particularly to the unserved sections of population. The International Institute for Communication and Development (IICD) has emphasized that ICT can be used to improve the quality of education by enhancing educational content development, supporting administrative processes in schools and other educational establishment and increasing access to education for both teachers and pupils via distance learning. As far as use of ICT in education is concerned, two concepts such as *e-learning* and *blended learning* are recently used to fortify the role of ICT in education. According to Tinio "e-learning encompasses learning at all levels, both formal and non-formal that uses an information network-the internet; an intranet (LAN) or extranet (WAN) – whether wholly or in part, of course delivery, interaction and/or facilitation". Others prefer the term on-line learning, web-based learning in a subset of e-learning and refer to learning by using an internet browser. The other term used in educational technology is blended learning. It refers to learning models that combine traditional classroom practice with e-learning solutions. For example, students in a traditional class can be assigned both print-based and online materials; have online mentoring sessions with their teachers through chats and subscribed to a class email list. UNESCO recognizes that "These

technologies have great potential for knowledge dissemination, effective learning, and the development of more efficient education services”.

In India satellite based teleconferencing (one-way video and two way audio) non-formal education has been operational since 1992 at national and regional levels. The ICT mission for school education in India aims to devise, catalyse, support and sustain ICT and ICT enabled activities and processes in order to improve access quality and efficiency in the school system.

According to UNESCO, ICTs can be used in education to:

- Improve administrative efficiency
- Disseminate teaching learning materials to teachers and students
- Improve the ICT skills of teachers and students
- Allow teachers and students access to sources of information from around the world
- Share ideas on education and learning
- Collaborate on joint prospects
- Conduct lessons from a remote location

The six focus areas of ICT in education programme as emphasized by the UNESCO are policy, teacher training, teaching and learning, non-formal education, monitoring and measuring, research and knowledge sharing. The international institutions as well as experts involved in communication and development believe that ICTs in education will be of great help to achieve the Millennium Development Goals of universal primary education through following ways:

- i) **Teacher Training:** Increase the supply of trained pre-service teachers through ICT-enhanced training and by creating teacher networks. Teacher training programme through ICT will be more effective and even can be conducted in a faster pace as compared to traditional methods.
- ii) **Teaching and learning in the class room:** The capacity development of teachers to empower them to use ICT in the classroom and the development of curricula and support materials/resources through ICT. The teaching- learning through ICT will not only improve the process but also raise the student attendance rate. It will enable both the teacher as well as students to use ICTs in their current as well as future teaching learning processes.
- iii) **Management and administration:** Improve the efficiency and effectiveness of Ministries of Education and related bodies through the use of ICTs for management and educational information. The department can use ICTs in procuring the record of the educational institutions, teacher and students and other lost and expenditure aspects relating to education. The on-line admission and on-line examination have fastened the educational administration and makes the administrative process cost-effective.

- iv) **Policy and strategy:** Establish an enabling environment and improve the overall strategic development of education by integrating ICT's policies and strategies into the education policies. The ICTs have also enabled the educational planners to make available educational policies, plans and strategies accessible to the teachers, students, administrators, and researchers. The feedback mechanisms have become easier with the advent of ICTs.

5.4.4 ICT and Health

The health care services will be improved and the quality increased by the coordinated induction of information technology in all segments of the health care action chain; local, regional and national health network will strengthen cooperation and resource management in the health care sector. According to the World Health Organisation (WHO) "Technology is the backbone of the services to prevent, diagnose and treat illness and disease. ICTs are only one category of the vast array of technologies that may be used. Given the right policies, organisation, resources and institution, ICTs can be powerful tools in the hands of those working to improve health".

According to WHO, the use of ICTs in the health sector is not merely about technology, but a means to reach a series of desired outcomes such as:

- i) Health workers making better treatment decisions;
- ii) Hospitals providing higher quality and safer care;
- iii) People making better choices about their own health;
- iv) Governments becoming more responsive to health needs;
- v) National and local information systems supporting the development of effective, efficient and equitable health systems;
- vi) Policy makers and the public aware of health risks and
- vii) People having better access to information and knowledge, they need for better health.

The use of ICT in health care has become an important aspect of health development. Health care solutions provided by ICTs are popularly known as *e-health*. The tools and services which contribute to e-health provide better and more efficient health care services for all. The use of e-health technologies allows a mutually beneficial collaboration and involvement of patients and medical professionals in the prevention and treatment of chronic diseases.

According to Dubow, ICT has produced impact on health care in following ways:

- i) Improved dissemination of public health information and facilitated public discourse and dialogue around major public health threats;
- ii) Enabled remote consultation, diagnosis and treatment through telemedicine;

- iii) Facilitated collaboration and cooperation among health workers, including sharing of learning and training approaches;
- iv) Supported more effective health research and the dissemination and access to research findings;
- v) Strengthened the ability to monitor the incidence of public health threats and respond in a more timely and effective manner;
- vi) Improved the efficiency of administrative systems in health care facilities.

In Peru, Egypt and Uganda effective use of ICTs has prevented avoidable maternal deaths. In South Africa, the use of mobile phones has enabled patients to receive timely reminders to take their medication. In Cambodia, Rwanda, South Africa and Nicaragua, multimedia communication programmes are increasing awareness of how to strengthen community response to HIV and AIDS.

Accurate, relevant and up-to-date information is essential to health service managers, if they are to recognize weakness in health service provisions and take action that will improve service delivery. Therefore, the development of effective information system is a necessary precursor to managerial improvement in health care system. The major elements of opportunities for ICT in primary health care are telemedicine and health services, health care data management, information systems, appropriate data collection devices and analysis tools, appropriate and affordable bio-medical equipment for grass root deployment, video/multi-model conferencing and e-connectivity and appropriate legal and administrative framework.

The ICT intervention in health development can be broadly categorised in to the following areas:

- i) **Telemedicine:** According to International Telecommunication Union, telemedicine is a powerful tool for empowering health care delivery. According to WHO, telemedicine is dealing of health care services where distance is a critical factor by health care professionals using information and communication technologies for the exchange of vital information for diagnosis, treatment and prevention of diseases and injuries, research and evaluation and for the continuing education of health care providers, all in the interest of advancing the health of individuals and their communities. The e-health is the use of emerging information and communication technology, especially the internet, to improve or enable health and health care.
- ii) **Health Management Information System (HMIS):** ICTs nowadays play an important role in most of the countries in maintaining medical records system, district, block and village health status records, disease surveillance system and health services delivery and recording system.
- iii) **E-learning and capacity building:** ICTs nowadays are being used for the capacity building of health and health related personnel by training institutions and NGOs involved in the health sector developmental

activities. The training organisations are providing access to internet and other materials to the trainees on recent development in health education.

- iv) **IEC on community-based health care delivery:** Information Education and Communication (IEC) is one of the important components of community health care strategies. Many studies have found that use of ICT materials in the community orientation training on health has better impact on knowledge, attitude and behaviour. Some of the organisations dealing with health sector development are laying emphasis on local information access and telemedicine with a local indigenous knowledge base.
- v) **Health Research:** The role of ICTs in health research is undoubtedly one of the important components of ICT in health development. The bibliographies in health give a number of useful references that focus on the utility of ICT in health research. It also emphasized on the use of electronic journals and on- line medical journals and the use of ICTs for networking among health researchers.
- vi) **Health Campaigns:** ICTs nowadays are being used in large scale health campaigns. Most of the African countries are using ICTs for the health campaigns of HIV/AIDS. USAID of South Africa seeks development of innovative public-private partnership that uses ICT to reduce the impact of HIV/AIDS in South Africa. The ICTs which are largely in use are mobile/ cellular technology, computer-based technology, radio, video television, web and social networking.

Media Lab Asia, 2005 recommendations on use of ICT in different levels of primary health care system are:

- i) The ICTs requirements for sub centre are handheld computer devices for data collection and compilation, the ICTs enabled digital camera, digital stethoscope, digital glucometer, portable digital weighting machine, e-mail facility, and multi-media facility for training.
- ii) The ICTs requirements for the PHCs are a PC with printer and web camera, digital weighing machine, digital stethoscope, digital ECG machine, digital glucometer, diagnostic test kits, pulse oximeter, digital x-ray, digital microscope, digital ultra sound etc.
- iii) ICTs requirements for Community Health Centre (CHC) will be all the requirements at PHC and diagnostic kits, cardiac monitor, ICT augmented operation theatre, good communication and connectivity facilities like email, fax and video conferencing and good telemedicine facilities.

It is rightly remarked that continuous development in ICTs has resulted in increasing use of those technologies in the practise of medicine and in the provision of medical care.

5.4.5 ICT and Women Empowerment

ICT can play a vital role in promoting women empowerment in developing

countries. There are many examples where women have used the new technologies to improve their business, create new business or find new employment opportunities. However, a large majority of women particularly those who are residing in rural and semi-urban areas are still excluded from the digital economy. Women need to become active promoters of the use of ICT and e-commerce and enhance their capacities in IT training and skills. Further, women also need to become more involved in ICT policy making and development to ensure that women's needs are incorporated in policies related to the infrastructure, access, training and education. ICTs can be a powerful catalyst for political and social empowerment of women. June Lennie (2002) citing the Australian case said that many women in rural Australia are taking leadership in community and economic development activities and are often extremely reliant on a range of communication technologies for personal, family, business and networking purpose. The UNESCO intends to overcome gender divide by providing opportunities to women in the knowledge societies through the development of ICTs. The UESCAP (United Nations Economic and Social Commission for Asia and Pacific) emphasised on closing gender divide through gender-responsive ICT capacity development for women's organisation and enhanced women's access to the benefits of ICT so that ICT becomes a central tool for women's empowerment and promotion of gender equality. The state of Kerala had initiated gender-focused approaches to ICTs and gender empowerment through Kudumbashree's ICT-based enterprise.

5.5 E-DEVELOPMENT AND ITS STRATEGIES

E-development is a process of development that makes use of ICTs or ICT applications to provide information and knowledge services necessary to enhance productivity, efficiency and quality of life. The International Institute for Communication and Development (IICD) considered e-development as that development by which 'e' means effective; 'e' means efficient and 'e' means empowering. It uses ICTs to increase peoples opportunities to empower poor people and to counter insecurity and vulnerability. ICT has a far reaching role to play in the development of a nation both urban as well as rural. Many of the developed nations have grasped the advantages of ICT which has contributed immensely to their socio-economic development. The e-development has occupied important place in the paradigm of development. Therefore, an efficient e-development strategy is *sine-qua-non* for development. It is emphasised that the measure of success of ICT in development will not focus on the spread of technology but on overall progress towards economic growth, and ultimately towards the Millennium Development Goals (Dutta et al., 2004).

Some of the strategies to be followed for e-development are as follows:

- i) **Raising awareness:** An intensive campaign in propagating the role of ICT in development is very much required. All the stakeholders at the national, regional, state and even at the grassroots level needed to be awarded about the role that ICT would play in development. It should clarify the ICT options available for development and those that should

be taken and used for development. The role of public, private sector, NGOs and even civil societies be harnessed which will go a long way to promote awareness on role of ICT in development.

- ii) **Building coalition:** e-development strategies in order to become successful need a coalition framework in which both the public and private sector should work shoulder to shoulder. A successful public-private-partnership model is very much required for the knowledge-based global economy.
- iii) **Clarifying roles and responsibilities:** A national strategy should help to clarify the roles and responsibilities of various actors of development such as government, private and civil society. The government need to take the lead by setting the policy and institutional environment in promoting ICT industry development. However, it should not be viewed as a government- only strategy.
- iv) **Scaling up:** Scaling up intra and inter ICT sectors is required. Therefore, reform and innovations are required to be made within the ICT sector. Besides, the intra-sector scaling up of the ICT to other development sectors like health, education, rural development, urban development, etc is also essential for the holistic development of a nation. Appropriate policy, institutional reforms and change in management practices are required during the scaling up commitment and knowledge about processes to diffuse and scale up best practices.
- v) **Leveraging ICT:** The role of e-development strategy can also be stated in terms of the three options for leveraging ICT such as an industry or sector in its own right, as a general purpose technology to be applied across sectors and an enabling infrastructure for empowerment and service delivery. To reduce the risks and improve the impact of targeting the ICT industry for promotion and focused efforts, governments should work with the private sector to identify target market opportunities, match specific niches to comparative advantage, systematically assess current constraints and jointly devise the policies and programmes to develop the industry and exploit market niches.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What do you mean by e-Development?

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2) ICTs are useful for development of Health? Comment.

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5.6 LET US SUM UP

ICT in this era of globalisation is considered as one of the important tools for development. Many of the developed countries have already reaped the benefits of ICTs and have successfully utilized it in their process of development. This unit describes in detail the meaning of ICT and its role in development of different sectors such as education, health, agriculture, industry, and poverty alleviation and women empowerment. Besides it also deals with the concept of development and various strategies of e-development.

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5.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) ICT refers to ‘information’, ‘communication’ and ‘technology’. Generally we seek information and communicate them to others through technology who may also be interested to know about its value. The ‘technology’ plays an important role in the effective transaction and communication of knowledge information to its seekers. ICT includes hardware, processes and systems that are used for storing and managing communications and sharing of information.
- 2) The Internet carries a vast array of information resources and services, most notably the inter-linked hypertext documents of the World Wide Web (WWW) and the infrastructure to support electronic mail, in addition to popular services such as video on demand, online shopping, online gaming, exchange of information from one-to-many or many-to-many by online chat, online social networking, online publishing, file transfer, file sharing and Voice over Internet Protocol (VoIP) or teleconferencing, telepresence person-to- person communication via voice and video.

Check Your Progress 2

1. ICT in order to promote development should be development centric. Asian Development Bank is helping its member countries in integrating ICT components in sector development strategies especially in education, health and agriculture. The focus is on improving public administration and finance management, as well as providing various electronic services to citizens and businesses. The ICT should provide information about performance of various sectors of development and different programmes of development.
2. ICT is effectively utilized for the development of rural areas at the grassroots levels by transacting the information about various developmental projects and programmes to the countryside masses.

Check Your Progress 3

1. The e-development is a process of development that makes use of ICTs or ICT applications to provide information and knowledge services necessary to enhance productivity efficiency and quality of life “The International Institute for Communication and Development (IICD), considered e- development is that development by which ‘e’ means effective; ‘e’ mean efficient and ‘e’ means empowering.
2. The use of ICT in health care has become an important aspect of health development. Health care solutions provided by information and communication technology (ICT) are popularly known as “e-health”. The tools and services which contribute to e-health provide better and more efficient health care services for all. The use of e-health technologies allows a mutually beneficial collaboration and involvement of patients and medical professionals in the prevention and treatment of chronic diseases.

UNIT 6 E-GOVERNANCE: POLICY AND FRAMEWORK

Structure

- 6.0 Introduction
- 6.1 Learning Outcomes
- 6.2 Concept of E-Governance
- 6.3 Stages of E-Governance
- 6.4 Models of E-Governance
- 6.5 Legal and Policy Framework
- 6.6 Significance of E-Governance
- 6.7 Challenges and Opportunities
- 6.8 Let Us Sum Up
- 6.9 Keywords
- 6.10 References and Further Readings
- 6.11 Check Your Progress: Possible Answers

6.0 INTRODUCTION

Reinventing government has been a dominant theme since the 1990s, wherein governments world over are attempting to improve the systems of public service delivery. Rapid strides made in the field of Information and Communication Technology (ICT) have facilitated the reinvention of governments and prepared them to serve the needs of a diverse society. In other words, the information age has redefined the fundamentals and transformed the institutions and mechanisms of service delivery forever. The vision is the articulation of a desire to transform the way government functions and the way it relates to its constituents. The concept of electronic governance, popularly called e-governance, is derived from this concern. Democracies in the world share a vision of the day when e-governance will become a way of life. India has been at the forefront of the IT revolution and has had its effect on the public administration systems, as we would see later in this Unit. If the potential of ICTs is harnessed properly, it has many opportunities, especially in the social and economic growth of the developing world.

6.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- discuss the concept and significance of e-governance;
- explain the various stages of e-governance;
- examine the various models of e-governance; and

- analyse the issues and challenges.

6.2 CONCEPT OF E-GOVERNANCE

E-governance is the application of ICT to the processes of government functioning for good governance. In other words, e-governance is the public sector's use of ICTs to improve information and service delivery, encourage citizen participation in decision-making and make government more accountable, transparent and efficient.

The Ministry of Information and Technology states that e-governance goes far beyond mere computerisation of stand-alone back-office operations. It implies fundamental changes in government operations; and a new set of responsibilities for the legislature, executive, judiciary and citizens.

According to the Comptroller and Auditor General, UK, e-governance means providing public access to information via the Internet by government departments and their agencies.

So in essence, e-governance is the application of ICT in government functioning to bring in **SMART** governance, implying: simple, moral, accountable, responsive and transparent governance.

Simple- meaning simplification of rules, regulations and processes of government through the use of ICTs and thereby providing for a user-friendly government.

Moral- connoting emergence of an entirely new system of ethical values in the political and administrative machinery. Technology interventions improve the efficiency of anti-corruption agencies, police, judiciary, etc.

Accountable- facilitating design, development and implementation of effective Management Information System and performance measurement mechanisms and thereby ensuring accountability of public service functionaries.

Responsive- streamlining the processes to speed up service delivery and make the system more responsive.

SMART governance, thus, helps in:

- improving the internal organisational processes of governments;
- providing better information and service delivery;
- increasing government transparency in order to reduce corruption;
- reinforcing political credibility and accountability; and
- promoting democratic practices through public participation and consultation.

E-governance and E-government

E-governance and e-government are often used interchangeably, so distinguishing between them at this stage is imperative. According to Thomas B. Riley government and governance are both about getting the consent and cooperation of the governed. However, whereas the government is the formal apparatus for this objective, governance is the outcome as experienced by those on the receiving end. E-government can be a more productive version of government in general if it is well implemented and managed. E-governance can evolve into participatory governance if it is well supported with relevant principles, objectives, programmes and architectures.

E-government is, thus, the modernisation of processes and functions of the government using the tools of ICT to transform the way it serves its constituents. As per the World Bank, e-government refers to the use by government agencies of information technologies (such as wide area networks, Internet and mobile computing) that have the ability to transform relations with citizens, businesses and other arms of government. It is the use of technology to enhance the access to and delivery of government services to benefit citizens, business partners and employees. E-governance, on the other hand, goes beyond the service delivery aspects and is seen as a decisional process. It is about the use of ICTs in the systems of governance that is, using ICT to involve multi-stakeholders in decision-making and in making governments open and accountable.

6.3 STAGES OF E-GOVERNANCE

Different stages of e-governance are identified on a specific set of criteria. These stages are:

- **Simple information dissemination** (one-way communication)- is considered as the most basic form, as it is used for merely disseminating information;
- **Two-way communication** (request and response)- is characterised with email system and information and data-transfer technologies in the form of the website;
- **Service and financial transactions**- is online services and financial transactions leading to web-based self-services;
- **Integration** (both vertical and horizontal)- in this stage the government would attempt inter and intra-governmental integration; and
- **Political participation**- this stage means online voting, online public forums and opinion surveys for more direct and broader interaction with the government.

Another classification of e-governance has six stages, of which the first two are similar to that of the above classification. The remaining four are:

1. **The third stage-** refers to multi-purpose portals, which allow customers to use a single point of entry to send and receive information and to process transactions across multiple departments;
2. **The fourth stage-** consists of portal personalisation, wherein customers are allowed to customise portals with their desired features;
3. **The fifth stage-** is when government departments cluster services along standard lines to accelerate the delivery of shared services and clustering of standard services; and
4. **Sixth and final stage-** technology is integrated further to bridge the gap between the front and back office.

After our discussion of the concept and stages of e-governance, we will now deal with significant models of e-governance that can be used in designing e-government initiatives.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit

1. Differentiate between E-governance and E-government?

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2. How many stages are involved in e-governance?

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6.4 MODELS OF E-GOVERNANCE

Prof. Dr. Arie Halachmi in his paper, namely, ' E-Government Theory and Practice: The Evidence from Tennessee (USA),' has given five essential models of e-governance, which can be used as a guide in designing e-government initiatives depending on the local situation and governance activities that are expected to be performed. These models are:

- The Broadcasting Model

- The Critical Flow Model
- The Comparative Analysis Model
- The E-Advocacy/Mobilisation and Lobbying Model
- The Interactive-Service Model

We will now discuss these models individually.

The Broadcasting Model

The model is based on dissemination/broadcasting of useful governance information, which is in the public domain into the wider public domain with ICT and convergent media. The strength of the model rests upon the fact that a more informed citizenry is better able to judge the functioning of existing governance mechanisms and make an informed opinion about them. Consequently, they become more empowered to exercise their rights and responsibilities. Widespread application of this model corrects ‘information failure situations’ by providing people with the relevant information relating to the governance sphere to make an informed opinion and impact governance processes.

Further, the use of ICT opens an alternative channel for people to access information as well as validate existing information from different sources.

The Critical Flow Model

The model is based on disseminating/channelling information of critical value to the targeted audience or into the wider public domain with ICT and convergent media.

The strength of this model is that ICT makes the concept of ‘distance’ and ‘time’ redundant when information is hosted on a digital network, and this could be used advantageously by instantly transferring the critical information to its strategic user group located anywhere or by making it freely available in the wider public domain.

The Comparative Analysis Model

This model is a highly significant model for developing countries and can be used for empowering people. Essentially, the model continuously assimilates best practices in the areas of governance and then uses them as benchmarks to evaluate other governance practices. It then uses the result to advocate positive changes or to influence ‘public’ opinion on these governance practices. The comparison could be made over a time scale to get a snapshot of the past and present situation or could be used to compare the effectiveness of an intervention by comparing two similar situations. The strength of this model lies in the infinite capacity of digital networks to store varied information and retrieve and transmit it instantly across all geographical and hierarchical barriers.

The E-Advocacy/Mobilisation and Lobbying Model

This model builds the momentum of real-world processes by adding the opinions and concerns expressed by virtual communities. This model helps the global civil society to impact on global decision-making processes. It is based on setting up a planned, directed flow of information to build strong virtual allies to complement actions in the real world. Virtual communities are formed which share similar values and concerns, and these communities, in turn, link up with or support real-life groups/activities for concerted action.

Hence, it creates a diversity of the virtual community, and the ideas, expertise and resources are accumulated through this virtual form of networking. Besides, it can mobilise and leverage human resources and information beyond geographical, institutional and bureaucratic barriers and use it for concerted action.

The Interactive-Service Model

It opens avenues for direct participation of individuals in governance processes and brings in greater objectivity and transparency in decision-making processes through ICT.

Fundamentally, ICT has the potential to bring in every individual in a digital network and enable interactively (two-way) flows of information among them.

Under this model, the various services offered by the government become directly available to its citizens in an interactive manner. It does so by opening up an interactive Government to Consumer to Government (G2C2G) channel in various aspects of governance, such as the election of government officials (e-ballots); redressing online of specific grievances; sharing of concerns and providing expertise; opinion polls on various issues; etc. (adapted from Prof. Dr. Arie Halachmi 'E-Government Theory and Practice: The Evidence from Tennessee, USA').

After our discussion about the models of e-governance, we will now focus on the legal and policy framework for the implementation of ICT and e-governance in the country.

6.5 LEGAL AND POLICY FRAMEWORK

The following provisions have laid down the legal and policy framework for ICT and e-governance.

Information Technology Act of 2000

The Action Plan endorsed by the Conference of Chief Ministers in 1987 had already addressed the pertinent issues of accountable and citizen-friendly administration; and transparency and right to information. In pursuance of these issues, the Information Technology Act was promulgated in 2000. The

objective of the Act is “to provide legal recognition for transactions carried out by means of electronic data interchange and other means of electronic communication, commonly referred to as ‘electronic methods of communication and storage of information’; to facilitate electronic filing of documents with the Government agencies; and further to amend the Indian Penal Code, the Indian Evidence Act, 1872, the Banker’s Book Evidence Act, 1891 and the Reserve Bank of India Act, 1934 and for matters connected therewith or incidental thereto.”

Both e-commerce and e-governance transactions are covered under the ambit of this Act, which facilitates acceptance of electronic records and digital signatures. The Act, thus, stipulates numerous provisions. It aims to provide for the legal framework so that legal sanctity is accorded to all electronic records and other activities carried out by electronic means. The said Act further states that unless otherwise agreed, an acceptance of the contract may be expressed by electronic means of communication and the same shall have legal validity and enforceability.

CHAPTER III of the Act details about 'Electronic Governance' and provides amongst others that where any law provides that information or any other matter shall be in writing or the typewritten or printed form, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied if such information or matter is:

1. rendered or made available in an electronic form; and
2. accessible to be usable for a subsequent reference.

Report of the Working Group on Convergence and E-governance 2002-07

Report of the Working Group on Convergence and E-governance proposed the need for the administration to transform itself from a passive information and service provider to a platform/ forum for the active involvement of citizens. This report primarily concerned itself with public investments. It could not visualise the extent of private initiative that could be expected to come forth in the convergence area or e-commerce or allied segments.

It felt the need to set up a central body for taking stock of the total IT picture in the country. This central body could be a ‘Council for E-governance’ or an ad hoc ‘Commission on Re-engineering Administrative Procedures for E-governance.’ Another alternative it suggested was to set up a National Institute of Smart Governance.

Common Minimum Programme:

The importance of e-governance has been recognised in the Common Minimum Programme of the UPA Government, which inter-alia states that e-governance will be promoted on a massive scale. It made a solemn pledge to the people of the country with a government that would be corruption free,

transparent and accountable; and an administration that would be responsible and responsive at all times.

National E-Governance Plan:

Three important elements of the National E-Governance Plan, which form the core infrastructure for effective service delivery are- Data Centres, State Wide Area Networks and Common Service Centres. The 10-point agenda of the Department of Information Technology announced for growth of ICT in the country includes expeditious implementation of a 'National E-Governance Plan' to bring about transparency and citizen-centric approach in administration.

Expert Committee:

An expert committee had also been constituted for the amendments in the IT Act 2000 to include the technological developments post IT Act 2000. The Expert Committee completed its deliberations and submitted its report in August 2005. Now the Expert Committee's recommendations have been put on the website of the Department of Information Technology for inviting public views and suggestions. The Committee, during its deliberations, analysed some of the relevant experiences and international best practices. The Committee, while formulating its recommendations, kept in view the twin objectives of (i) using IT as a tool for socio-economic development and employment generation; and (ii) further consolidation of India's position as a significant global player in the IT sector.

Right to Information Act 2005

The Right to Information Act 2005 confers on the citizens the right to:

1. inspect works, documents and records of the government and its agencies;
2. take notes, extracts or certified copies of documents or records;
3. take certified samples of material; and
4. obtain information in the form of printouts, diskettes, floppies, tapes, video cassettes or in any other electronic mode.

This has ensured a transparent and accountable government to the people. It has also established a two-way dialogue between the citizens and the government. It has enabled citizens to make well-informed decisions. Further, it is an essential step towards tackling corruption and has ensured better monitoring of services provided by the government.

6.6 SIGNIFICANCE OF E-GOVERNANCE

ICT applications impact the structures of public administration systems. Technological advancements facilitate administrative systems by enabling:

- Administrative Development; and

- Effective Service Delivery

We will now discuss them individually.

Administrative Development

Administrative reforms, often, have focused on procedural details and restructuring of systems and processes of government organisations. The primary objective of these reforms is to enhance the capacities of the systems. ICTs can be used and are being used now to give further impetus to the process. They help in the following manners:

Automation of Administrative Processes

A truly e-governed system would require minimal human intervention and would rather be system driven. While initially, the solutions that were offered were relatively primitive with poor information layout, inadequate navigation provisions, occasional disruption in services, periodic outdated content and little or no 'back office' support. However, technological advancements and increased pressure from citizenry have prompted improvements in these areas. Now administrative departments are computerised and connected through networks. The software has been built and designed around government departments ensuring efficiency in operations. The departments have launched individual websites carrying information of their respective departments. This has enabled online carrying of operations and file movements. Budgeting accounting, data flow, etc. has become easy. This has increased the efficiency of office operations and processes and has reduced unnecessary delays.

Paperwork Reduction

An immediate impact of automation would be on the paperwork. Paperwork is reduced to a greater extent with communication being enabled via electronic route and storage and retrieval of information in the electronic form. All this has led to the emergence of 'less paper office'. This concept is defined as an office situation where all the information (file and mail) amongst various functionaries is distributed online. In the words of Dubey, less paper office is the implementation of effective electronic communication processes that enable elimination of reproductive works and unnecessary papers. The concept is where files and emails (information) are transmitted over wires to small computers at each employee's desk. Office work, such as file movements, notings, etc. is computerised, and documentation, report preparation, databases are now maintained in computers. Due to interconnectivity through LAN, transfer of information and files take place online, thus reducing the physical movements and consumption and storage of vast piles of paper.

Quality of Services

ICT helps governments to deliver services to the citizens with greater accountability, responsiveness and sensitivity. Quality of services improves,

as now the people can get services efficiently and instantaneously. As volumes of transactions and information can be electronically handled and delivered over a wider area through the net and web, qualitative services become possible in the least time, in the least cost, in the least difficulty and greater convenience.

By ensuring online redressal of grievances, the accountability of officials is ensured. They have become sensitive to the issues affecting people. Monitoring by way of video teleconferencing has further facilitated central monitoring, reporting and face to face communication that has assured effective service delivery by the officials.

Elimination of Hierarchy

ICT has reduced procedural delays caused by hierarchical processes in the organisation. Through Intranet and LAN, it has become possible to send information and data across various levels in the organisation at the same time. Computerisation and communication patterns facilitated by ICT have increased efficiency and have led to the involvement of all levels in decision-making.

Change in Administrative Culture

Bureaucratic structures have been plagued by characteristics aptly described by Victor Thompson as 'bureau-pathology'. From the days of New Public Administration, efforts have been made to find ways to deal with the pathological or dysfunctional aspects of bureaucratic behaviour and to make delivery of public services effective and efficient. With e-governance, public actions coming under public glare would certainly induce norms and values of accountability, openness, integrity, fairness, equity, responsibility and justice in the organisational culture. Instead, the administration would become efficient and responsive.

Effective Service Delivery

ICTs play an essential role in effectively delivering services to people. ICTs ensure:

- **Transparency** by dissemination and publication of information on the web. This provides easy access to information and subsequently makes the system publicly accountable. Also, as the web enables the free flow of information, it can be easily accessed by all without any discrimination.

Economic Development

The deployment of ICTs reduces transaction costs, which makes services cheaper. For example, rural areas suffer on account of lack of information regarding markets, products, agriculture, health, education, weather, etc. and

if all this could be accessed online would lead to better and more opportunities and thereby prosperity in these areas.

Social Development

The access to information empowers the citizens. Informed citizens can participate and voice their concerns, which can be accommodated in the programme/ project formulation, implementation, monitoring and service delivery. Web-enabled participation will counter the discriminatory factors affecting our societal behaviour.

Strategic Information System

Changing the organisational environment and increasing competitiveness have put pressures on the performance of the functionaries. Information regarding all aspects needs to be made available to the management at every point to make the routine as well as strategic decisions. ICTs effectively enable putting such strategic information systems in place.

After the discussion, as mentioned above, on the significance of ICTs in governance, we will now highlight specific measures that will enable its effective implementation.

6.7 CHALLENGES AND OPPORTUNITIES

The above discussion highlighted the critical role of ICTs in governance. In order to harness the benefits of ICTs maximally, we need to develop sufficient and adequate infrastructure, provide sufficient capital and investment, enable easy and more comprehensive accessibility and generate ample and skilful human resources. These are some of the immediate and pertinent challenges to effective implementation of ICT and e-governance. We will now discuss these issues individually.

Infrastructure:

The foundation of e-governance is based on telecommunication services. To develop telecommunication, infrastructures are to be created so that the end-user can access the services promptly and effectively. To strengthen the infrastructure, 'The National Task Force on Information Technology and Software Development' in 1998 recommended broadband connection (also known as 'the last mile') linkage for IT Applications Service Providers (ASPs), Internet Service Providers (ISPs) and IT promotional organisations, either by fibre optics or by radio communication, intending to 'boost efficiency and enhance market integration' through Internet/Intranet for sustainable regional development.

Capital:

A high rate of investment in IT capital and a supportive environment is necessary to achieve a digital economy. Given the resource crunch with the

government, there is a need to generate resources from the market and private sector. The public-private partnership may be beneficial in this regard, as the private sector can participate and contribute to capital and expertise support.

Access:

At present, there are more than 10 million users of the Internet in the country. However, the irony is that more than 75 per cent of these users are in urban India. The Internet has still to reach the rural and disadvantaged sections. However, efforts are being made to expand ICT connectivity into rural areas through the involvement of Gram Panchayats. NIC has developed a comprehensive web-based software for Panchayati raj and rural applications, which is being implemented in states like Andhra Pradesh. With most of the panchayats getting computerised, accessibility to various services has become easy.

The utility of information:

There is a need to provide information, which is useful. The content of the information should be such that it should be attractive, beneficial and appealing to the people. In this regard, the Government of India and some of the state governments have prepared a vision document for e-governance keeping in mind the needs of the citizens. Though Citizens' Charters of many departments are available on the net, further publicity of such facilities is required to enable the public to access the necessary information.

Human Resource Development:

Despite the ascending growth rate observed in employment in the IT sector, there is a dearth of quality human resources. There exist a demand and supply gap in the IT human resources market. India needs to have more technical institutes to impart education and training to build a pool of human resources in the field.

Capacity Building:

Service delivery will be useful if there are trained human resources. Though computer training is being imparted to all the necessary public functionaries, except in a few cases, effective use of ICT is yet to be seen. Moreover, there is an immediate need to launch a nationwide 'Train the Teachers Programme' (3T Programme). This should be done at all levels, including schools and colleges. A combination of physical and virtual training also needs to be imparted.

Changing the Mindset of Government Functionaries:

To accept the change, there is a need to change the mindset of service providers and receivers. The government functionaries need to be made aware that they are there to serve the clients as per the policies and programmes and that technological advancement is only a facilitator to

solutions of problems faced by people and not a solution in itself. To change the mindset of the service providers there is a need to impart orientation and training programmes to them.

Language:

The success of e-government also depends on communication with the people in their local languages. Currently, the most widely used language is English for e-government. Nevertheless, given the Indian social conditions, unless we develop interfaces in vernacular languages, it would remain out of reach of many people who are not capable of accessing these services in English. In this context, it is essential that a clear strategy is formulated to provide access to local level databases maintained in regional and local languages as well as to use appropriate interfaces to aggregate such data. However, it may be mentioned here that organisations like the Centre for Development of Advanced Computing (CDAC) have developed multilingual software for the purpose.

Standardisation in Data Encoding:

Once multiple access points maintained in various languages at various levels are established, there is a need to update them in conformity with similar standards for data encoding-an application logic for a common horizontal application and data dictionary. This is also important for finding aggregates in the national context.

Grievance Redressal Mechanism:

The mechanism planned for various functions needs to make provision for grievance redressal as well. Interactive platforms on the Internet may speed up the process and may be useful in this regard. The BMC-Praja Foundation's joint initiative of the Online Complaint Management System (OCMS) is perhaps the world's first in citizen-government partnership for solving public grievances in municipal services. Inaugurated in April 2003, the OCMS has been receiving grievances on behalf of citizens availing services of municipalities in Mumbai. It uses IT as a tool to bring in efficiency and effectiveness into the system. One can register his/her complaint online regarding various municipal services, and the Municipal Corporation will redress this complaint in the time stipulated in the Citizens' Charter adopted by the Corporation.

The Central Vigilance Commission has also provided such a platform for people to register their complaints against corrupt officials. Such sporadic instances need to be made broad-based and effective, though it may be conceded that more and more public service agencies are now providing or contemplating such facilities.

Cyber Laws:

The government needs to enact appropriate laws, especially those, which are necessary to enable transactions over the Internet. Safety concerns regarding the use of credit cards or other modes of payment stop the consumers from using such facilities. The Mahanagar Telephone Nagar Limited, Delhi, for example, has provided the online facility for payment of telephone bills. However, not even one per cent of its consumers are making use of this provision. Hence, security has to be ensured for generating confidence in the system.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit

1. Explain the benefits of e-governance in increasing quality of services?

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2. Briefly explain the main challenges and opportunities of e-governance?

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6.8 LET US SUM UP

According to Traunmuller and Lenk, e-governance is a global phenomenon today, and it is the most recent paradigm in public administration. The speed and transparency associated with e-governance have the potential to make public administration responsive and effective. As the development of e-governance gets past the phase of pilot projects, it becomes apparent that sustainable development of e-governance will depend on an adequate institutional framework that will enable public administration to manage and harmonise the emerging multitude of technical and organisational changes at all levels of government....

The time has come to focus on the challenges in implementation, especially those related to cross-level applications and institutional framework, which would enable to bring in broader changes in governance.

6.9 KEYWORDS

E-governance: use of technology to enhance the access to and delivery of government services to benefit citizens, business partners and employees.

ICTs: are the information and communication technologies such as radio, computers, Internet, Intranet, Websites, and satellites; providing database, knowledge database, expert systems, Geographic Information System, Management Information System, video and audio teleconferencing.

State Wide: networks are linking the state headquarters right up to the block Area Network level through National Informatics Centre Network.

National E-governance: seeks to implement 25 Mission Mode Projects at the Plan Centre, State and integrated service levels to create a citizen-centric and business-centric environment for governance, create governance and institutional mechanisms, set up core infrastructure, formulate key policies and channelise private sector technical and financial resources into the national e-governance efforts.

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6.11 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. E-governance and e-government are often used interchangeably, so distinguishing between them at this stage is imperative. According to Thomas B. Riley government and governance are both about getting the consent and cooperation of the governed. However, whereas the government is the formal apparatus for this objective, governance is the outcome as experienced by those on the receiving end. E-government can be a more productive version of government in general if it is well implemented and managed. E-governance can evolve into participatory governance if it is well supported with relevant principles, objectives, programmes and architectures.
2. There are six stages of e-governance. Primarily, these stages are associated with
 - **Simple information dissemination** (one-way communication)- is considered as the most basic form, as it is used for merely disseminating information;
 - **Two-way communication** (request and response)- is characterised with email system and information and data-transfer technologies in the form of the website;
 - **Service and financial transactions**- is online services and financial transactions leading to web-based self-services;
 - **Integration** (both vertical and horizontal)- in this stage the government would attempt inter and intra-governmental integration; and
 - **Political participation**- this stage means online voting, online public forums and opinion surveys for more direct and broader interaction with the government.

Check Your Progress: 2

1. ICT helps governments to deliver services to the citizens with greater accountability, responsiveness and sensitivity. Quality of services

improves, as now the people can get services efficiently and instantaneously. As volumes of transactions and information can be electronically handled and delivered over a wider area through the net and web, qualitative services become possible in the least time, in the least cost, in the least difficulty and greater convenience. By ensuring online redressal of grievances, the accountability of officials is ensured. They have become sensitive to the issues affecting people. Monitoring by way of video teleconferencing has further facilitated central monitoring, reporting and face to face communication that has assured effective service delivery by the officials.

2. In order to harness the benefits of ICTs maximally, we need to develop sufficient and adequate infrastructure, provide sufficient capital and investment, enable easy and more comprehensive accessibility and generate ample and skilful human resources. These are some of the immediate and pertinent challenges to effective implementation of ICT and e-governance.



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UNIT 7 E- GOVERNANCE IN RURAL DEVELOPMENT

Structure

- 7.0 Introduction
- 7.1 Learning Outcomes
- 7.2 Meaning and Importance of e-Governance
- 7.3 E-Governance and Rural Development
- 7.4 Dimensions of Digital Divide
 - 7.4.1 Economic Dimensions of Digital Divide
 - 7.4.2 Political Dimensions of the Digital Dimensions of the Digital Divide
 - 7.4.3 Spatial Dimensions of the Digital Divide
 - 7.4.4 Social Dimensions of the Digital Divide
 - 7.4.5 Cultural Dimensions of the Digital Divide
- 7.5 Models of e-Governance in Rural Development
 - 7.5.1 Broadcasting/Wider Dissemination Model
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 - 7.5.5 Interactive-Service Model/Government-to-citizen-to-Government Model (G2C2G)
 - 7.5.6 E-Governance and Maturity Models
- 7.6 Cases in Rural e-Governance in India
- 7.7 Let Us Sum Up
- 7.8 References and Further Readings
- 7.9 Check Your Progress: Possible Answer

7.0 INTRODUCTION

Use of Information and Communication Technologies (ICTs) has traversed a long way and in the present context, it has in fact become a way of life. ICT in its nascent stage remained with researchers and academicians. Later, its usefulness was leveraged for the benefit of the business sector, especially to support the economics of products and services with specific focus on optimization of transaction and coordination costs. Use of ICT in public sectors has now been for more than five decades. Today, prefixing “e” with various processes (business, government, development etc.) has not remained as statutory, but has transformed them as artefacts for real use.

The United Nations Development Programme (UNDP) study for Asia acknowledges the difficulty in appreciating the role of technology in achieving SDGs, especially in addressing those issues related to poverty reduction, health and primary education. It infers that growth of ICT in Asia has not been a critical factor in progress towards the attainment of SDG. The

study also indicates statistically not significant correlations in the use of ICT and improvements in SDG indicators and strongly argues in favour of the role of ICT in addressing issues related to information asymmetry. In developed countries, SDG indicators are strongly correlated with ICT interventions. It is experienced that ICT infusion would largely depend on the demand for services and efforts should be made to generate such demand by making ICT affordable, accessible and usable.

7.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- Explain the meaning and importance of e-governance in rural development;
- Discuss the dimensions of digital divide;
- Explain the models of e-governance; and
- Describe the perspectives of e-governance with examples.

7.2 MEANING AND IMPORTANCE OF E-GOVERNANCE

While embarking on a discussion on e-governance, it is essential to understand the spheres of *e-government* and *e-governance* systems that exist in the domain of public services. E-government is broadly defined as an effort to transform government services with focus on information based services for citizens. It also aims to provide ICT enabled environment in government departments to improve public services and in turn support the democratic process. E-government is also categorised to be ICT-enabled transformed government process to develop and deliver high quality seamless and integrated public services, to enable effective constituent relationship management; and to support the economic and social development goals of citizens, business and civil society at local, state, national and international level. E-government is defined as an ICT enabled SMART government system. It is also aimed to interface with various Government systems (G2G), Businesses (G2B), Citizens (G2C) etc. In summary, e-government is all about ICT enabled government institutions/processes and agencies.

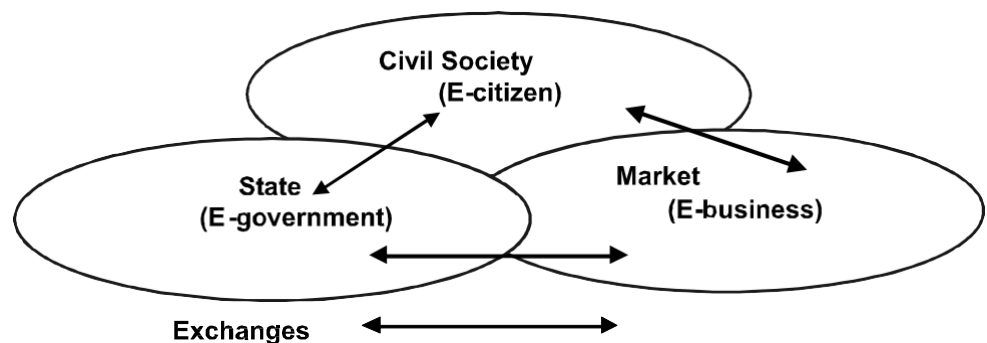


Fig. 6.1: E-Governance Scenario

E-governance is primarily a transformed version of *governance* through ICT intervention. Therefore, it is expected that ICT applications need to maintain the sanctity of governance in all its forms, content and spirit. E-governance gets the support of e-government systems that operate as a back-end service provisioning entity to coordinate and extend support to other agencies working in tandem for reinforcement of public service deliveries. In Fig. 6.1, scenario for e-governance is presented. It may be seen here that e-governance encompasses the rigour of e- government systems, provides the desired structure for networking among market (e-business) and civil society (e-citizen). This relationship indicates that e- governance provides support to the policy makers, decision makers and other stakeholders to achieve the goals set for overall development of society in general. E-governance advocates e-democracy where all interactions among electorate and the elected are performed electronically. There are various e-governance and e-government models and frameworks developed to understand the complexity of the issue, to appreciate the areas of ICT interventions and to make clear the deliverables through the ICT enabled processes.

Key advantages of e-Governance

- Improve service delivery mechanism;
- Enhancing the efficiency in output;
- Emphasizing on wider access to information;
- Better management of accounting and record keeping;
- Promote accountability and transparency.

7.3 E-GOVERNANCE AND RURAL DEVELOPMENT

Pre-fixing the letter *e* to *governance* and *government* is a challenging phenomenon for planners, administrators and political systems globally. In the context of rural development, the situation is more challenging because of digital divides.

The digital divide, commonly understood as the gap between ICT *haves* and *have-nots*, has emerged as an important issue of our times largely due to the uneven diffusion of the Internet. We all share the same belief that the use of ICT for development holds very important promises. The belief sometimes becomes myth or magic... “for instance when one thinks that the mere fact of connecting a person to the Internet is going automatically to initiate a process causing this person to escape his/her situation of poverty. Many also argue that mere possession or use of ICT devices does not exclude a person from the ambit of digital divides. It is said that digital divide has many dimensions to understand it holistically. Digital divide is considered a “devil” in disguise and can affect an individual, organisation and even a country”.

In the year 2000, the G8 initiated the *Digital Opportunity Task Force* (Dot Force) to raise international awareness on the subject. Since then a concept has prevailed to fight against the digital divide as a priority because ICT offers many possibilities for development of people, communities and

countries. The concept of ICT for development (ICT4D) is now used by many international, regional and national organisations, and by all sectors (international, governmental, corporate, civil society, and the academy) to address this phenomenon.

Analyzing the digital divide becomes important for two reasons. The first is that if the digital divide is bridged, it may contribute to economic growth. The second and the more important reason is that the digital divide remains one of the most patent examples of an iniquitous fact of the current trajectory of global development. So, bridging this digital divide becomes an economic as well as a moral imperative.

7.4 DIMENSIONS OF DIGITAL DIVIDE

Digital divide is a complex phenomenon and does not limit itself to provision of ICT. There are various dimensions to this divide and a holistic view is necessary to understand its complexity. Relevant dimensions of digital divide discussed here include *economic, political, spatial, social and cultural*.

7.4.1 Economic Dimensions of Digital Divide

The economic dimensions of the digital divide are shown in the Fig. 6.2. It is essential to understand this dimension since ICT use has been largely influenced by *micro-economic theory* and *transaction cost theory* and *agency cost theor*".

- i) **Micro-economic Theory (Micro Economics):** The defining feature of the new economy is the increasing optimization of costs of variety of goods and services being created for the individual and society. Information is being considered as an important *factor of production* in all the areas related to production of goods and services. Information enabled production, distribution and retailing systems can influence the marginal costs in all these areas. Knowledge is also considered a very important dimension of the economy. Knowledge goods are subject to increasing economies of scale. But it tends to create monopoly because the low marginal cost means that firms can always lower their prices temporarily to eliminate new competitors. Use of knowledge goods without an attendant can increase in access which may mean that the digital divide may increase to infinite proportions.
- ii) **Transaction Cost Theory (Transaction Economics):** A transaction is termed as an instrument for exchange of goods, services and information. ICT helps in optimizing this cost because of its inherent capabilities to extend the services rendered by the service providers at the door steps of the service users. Due to this ICT enabled service orientation, digital divide can be addressed successfully because of the fact that transactions would happen locally with the ownership lying with the users. This empowerment with information sharing would help reduce the cost of information asymmetry.

- iii) **Agency Cost Theory (Agency Economics):** Agents in the economic processes play a vital role in controlling the business. They not only are able to create information asymmetry, but also add cost to the goods and services being rendered. Virtualization of organizations, introduction of ICT enabled networks and creating ICT enabled service user interfaces provide the ambience to reduce the agency costs and increase service quality. Extent of availability of ICT infrastructure determines gravity of this dimension of digital divide since networking influences agency cost considerably.
- iv) **Entry barriers:** ICT mediated virtual organisations have the advantages of establishing networks for e-business (B2C, B2B, C2C etc). These organisations generate the capacity to leverage information and knowledge on market, customers and consumer behaviour leading to creation of market oriented products and services. Such capabilities may lead to digital divide among various organisations that do not have these capabilities to enter into new areas and manage competition.

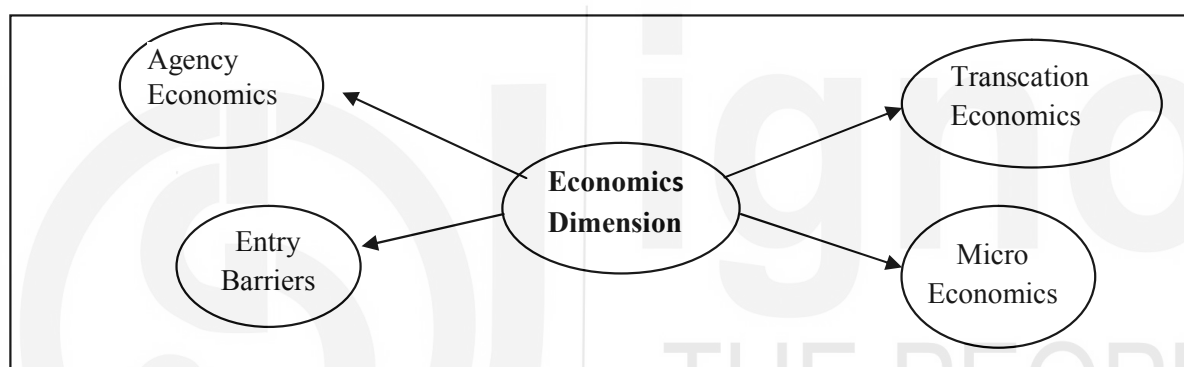


Fig. 6.2: Economic Dimensions of Digital Divide

7.4.2 Political Dimensions of the Digital Divide

Digital divide is largely and adversely influenced by this dimension since political will and short sighted compulsions have far reaching effects in establishing the right infrastructure and its judicious use. This dimension has 3 components as shown in Fig. 6.3.

- i. **Government Regulation:** Evolution of ICT infrastructure use has roots to prolific advancements in internet technologies and supporting infrastructure encompassing all the dimensions of ICT. Internet has remained a monopoly of national and international communities. Various regulating agencies have emerged to control the availability, rights to use and own ICT services across the globe. Almost all the countries today have strategies for e-governance and deployment of spectrum for common use. There is a visible shift from highly regulated environment for effective use of ICT services to its democratic use. However, many countries (especially developing ones) still face mounting challenges for having free access to the ICT infrastructure, bandwidth and all related services. Political consortium like Euro, G8 etc., play crucial roles in controlling these services and thus, leading digital divides at the international level.

- ii. **ICT Justice:** The constraints to internet access are not only economic but also political, given that the electronic dissemination of knowledge, by making people aware of their rights and by disseminating news and views about the injustices of society can challenge established relations of power. This may mean that the provision of internet facilities may not be in the interest of the dominant powers. Theoretically, the internet can enable everyone to participate in the political process with equal rights to access. Further, it can give a political voice to the numerous marginalised groups across the world by allowing them to network with each other and by broadcasting their plight to the rest of the world. Today right to information (RTA - Right to Information Act) is providing such an ambience, thus leading to digital opportunities in India.
- iii. **Enhancing Democracy:** The internet can also qualitatively quicken up the different political processes like elections, referenda, recall etc. Thus, the internet has an enormous democratic potential. “While a relatively small, educated, and affluent elite in a few countries and cities would have access to an extraordinary tool for accessing information and for political participation, actually enhancing their citizenship, the uneducated, switched off masses of the world, and of the country, would remain excluded from the new democratic core, as were slaves and barbarians at the onset of democracy in classical Greece”.

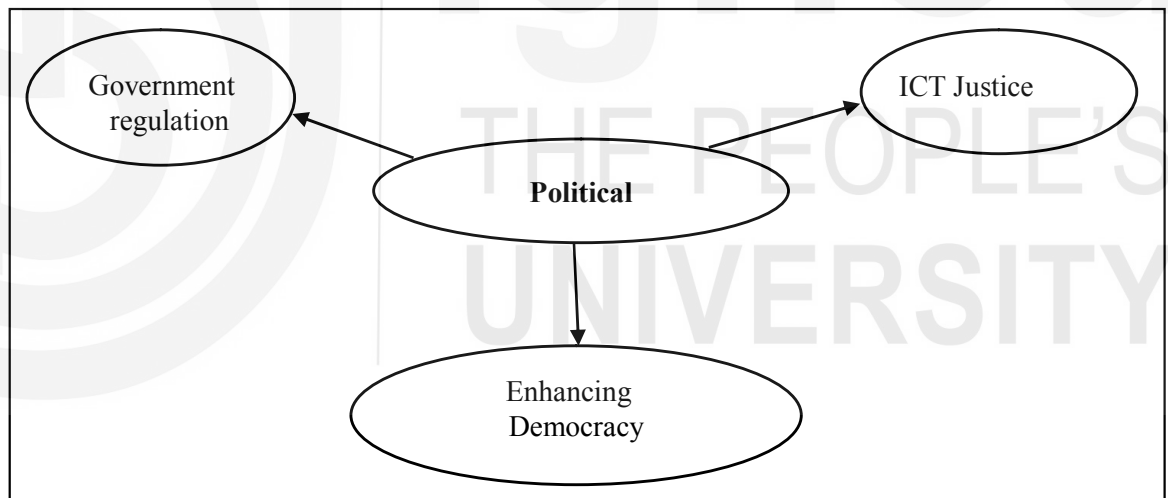


Fig. 6.3: Political Dimensions of Digital Divide

7.4.3 Spatial Dimensions of the Digital Divide

This is a critical dimension of digital divide as it relates to its transaction economics dimensions (Fig. 6.4):

- i. **Rural-Urban:** It is a common notion that rich, influential classes in the society cutting across the developing countries have easy access to ICTs. Evolution of internet suggests that since its inception at the hands of the US military, the very architecture of the internet has revolved around a handful of nodes that route internet traffic, all of which have been clustered in cities of academic or governmental significance. The resulting patterns of service provision became steadily restructured by

corporate Internet Service Providers (ISPs) in partnership with backbone providers generating geography centred largely in urban areas, whose concentrations of affluent users generate economies of scale that lead to the highest rates of profit. This widens the rural-urban divide. However, recent trend shows an improvement with the advent of deregulation, globalization and restructuring of development economics.

- ii. **Concentration of Innovation Centres in Metropolitan Areas:** It is widely believed that ICT firms have concentrated in urban areas. These firms have become hubs for knowledge workers whose aim is to innovate and provide improved services. These firms have been successful in attracting people to these urban areas rather than by developing local skills. Thus, this disparity has contributed to the digital divide.
- iii. **Concentration of Quality of Services:** There is a hype that cyberspace spells the end of geography because of the instantaneous communication that it can offer to its users. It is also believed that cyberspace also promises universal and democratic entry to the electronic highways of the world economy. But contrary to these claims, access to the internet is very unevenly distributed both socially and spatially. This seems to be the trend in the third world countries also. The implications could be that the quality of services will be different in different geographical areas.

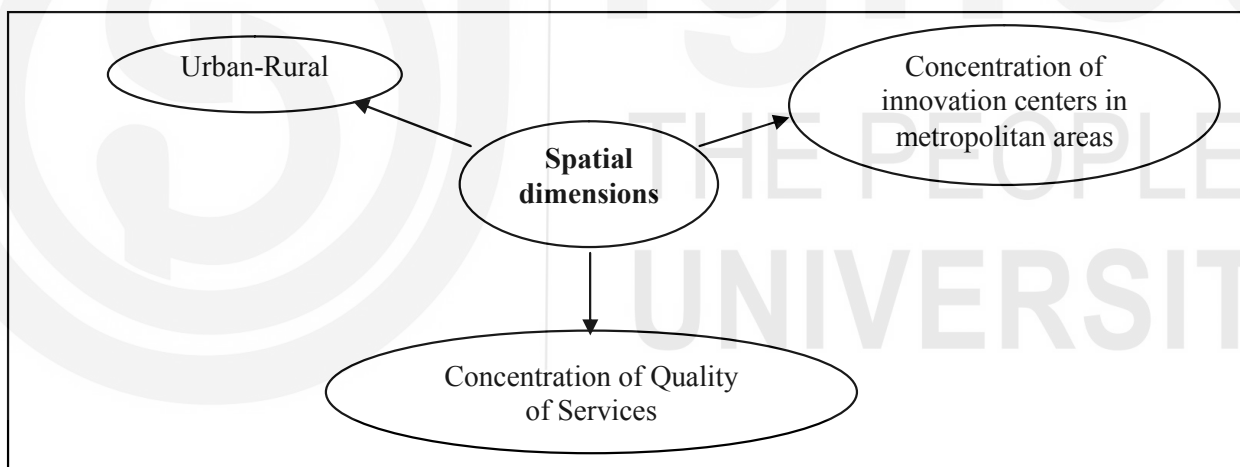


Fig. 6.4: Spatial Dimensions of Digital Divide

7.4.4 Social Dimensions of the Digital Divide

Digital divide largely is influenced by the social dimensions and especially developing countries are adversely affected by this (Fig. 6.5).

- i. **Income Groups:** Scholars have noted that the internet tends to reinforce existing class and social relations both within and across countries. The digital divide does not create new fault lines in society, but by and large replicates the existing social stratification (at least when defined in terms of physical access). Further, certain sectors and societies are completely excluded from the new networks of information, effectively creating black holes of informational capitalism. These black holes constitute the

Fourth World, which comprises large areas of the globe, such as sub-Saharan Africa, impoverished rural areas of Latin America and Asia. So, in a broad sense, we can assume that income affects the access to ICTs.

- ii. **Gender Bias:** Internet use is heavily gendered. In a patriarchal culture, the socialization of girls does not make them friendly to technology. To them, technology is said to be a guy thing and the internet is no exception to this rule. So, they are frequently under-confident when working with machines.
- iii. **Information Asymmetry:** A recent paper from demos (a UK based think- tank) argues that the real challenge is not exclusion from information, but exclusion by information. If there is information about a person, then the dominant powers can decide whether to include or exclude that person in other networks of power, influence or wealth. “The rise of a digital economy based on detailed personal profiles means that organizations – public, private and voluntary (NGOs) – are increasingly able to exclude large numbers of people from access to basic services and opportunities on the grounds that they constitute ‘bad risks’.
- iv. **Theory-Practice Gap:** “The digital divide – the gap between those sections of the population with and those without access to ICT is a highly debated social issue”. But there seems to be more of intellectual debates and discussions going on about the digital divide than substantive efforts being taken to bridge the divide. We are currently in a phase of history which is unprecedented in the levels of economic inequality seen. So, as a society the collective conscience as Emile Durkheim would say, has to be satisfied that there is no grave injustice being done and that everything is fine. This is needed because only then can society function as it is. So, the huge amount of discourse about the need for bridging the digital divide may serve this purpose of a self-fulfilling legitimizing mechanism which helps to maintain a healthy collective conscience at the national and the global realm, which helps to legitimize the status-quo.

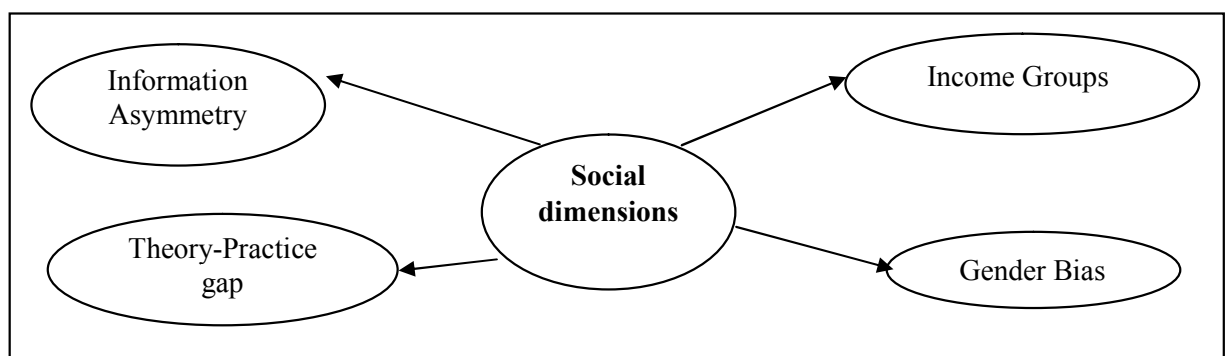


Fig. 6.5: Social Dimensions of Digital Divide

7.4.5 Cultural Dimensions of the Digital Divide

The cultural dimensions of digital divide are shown in Fig. 6.6:

- i) **Computer Human Interfaces (Language):** Language is a definite barrier since roughly 60% of all internet content today is in English. So, one cultural dimension of the divide is currently manifested as a divide between the English and the non-English speakers. But for those huge regions that share common languages and thus have big linguistic groups like the Arab world and much of Latin America – the internet can be the great unifier and creator of business opportunities and hence, of higher living standards (Cattai, 2001). But a digital divide may potentially emerge between these big linguistic groups and the rest of the small linguistic groups which don't have such a common critical cultural mass.
- ii) **Cultural Bias for ICT adoption:** There is a concept called Cultural Lag by David Ogburn. It says that the Culture always lags behind technological innovations. So, even if ICT is implemented, the mindset of the users may not be ready to welcome it. This is widely seen in the lower levels of the government bureaucracy.

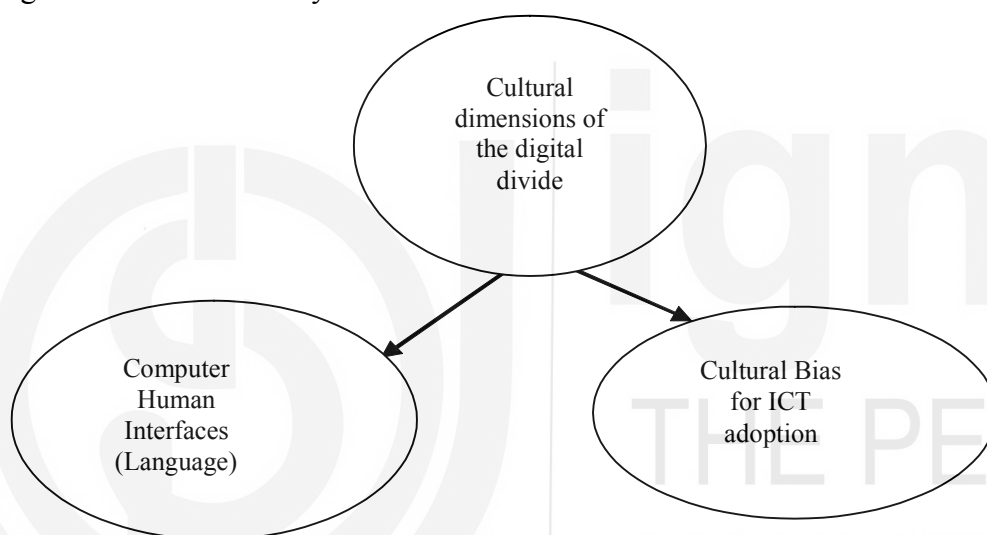


Fig. 7.6: Cultural Dimensions of Digital Divide

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit

1. Enumerate the key advantages of e-Governance?

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2. What are the five dimensions of the digital divide?

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3. With reference to spatial dimensions of the digital divide, compare the urban with rural telecom scenario.

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7.5 MODELS OF E-GOVERNANCE IN RURAL DEVELOPMENT

Models in various forms are being tried to showcase ICT interventions for development which would benefit rural mass. There have been many government and non-government agencies working either disjoint or collectively for this cause. E-governance is often being contemplated as one of the major deliveries through ICT interventions. But most of the ICT interventions (either in pilots or in actual implementation) in rural areas are facing challenges of inadequate supporting infrastructure like electricity.

7.5.1 Broadcasting/Wider Dissemination Model

The *Broadcasting and Dissemination model* is based on dissemination of information relevant to better governance that is already in the public domain. It includes use of ICT and convergent media for wider reach. The rationale behind this model is that a more informed citizenry is able to understand better the governance mechanisms and is more empowered to make informed choices and exercise its rights and responsibilities. Further, there is a greater likelihood that the society in which the individuals are equally informed will ensure that the agenda and forms of governance are not biased to favour a few. The wider dissemination model opens up an alternative channel for people to access information as well as to validate information available in the local domain from external sources. The widespread application of this model gradually corrects the situation of information failure and provides people with the basic government-related information to come to a common understanding and decide upon the future course of action.

7.5.2 Critical Flow Model

This model is based on channelling information of critical value to a targeted audience or spreading it in the wider public domain through the use of ICT and convergent media. It requires foresight to understand the significance of a particular information set and use it strategically. It may also involve locating users to whom the availability of a particular information set would make a critical difference in initiating good governance. The strength of Critical Flow Model is the inherent characteristic of ICT that makes the notion of distance and time redundant. This reduces the cases of exploitative governance possible earlier due to time lag between availability of information to different users.

7.5.3 Comparative Analysis Model

The Comparative Analysis Model is based on exploring information available in the public or private domain and comparing it with the actual known information sets to derive strategic learnings and arguments. The model continuously assimilates new knowledge products and uses them as a benchmark to evaluate, influence or advocate changes in current governance policies and actions. The comparison could be made over a time scale to get a snapshot of the past and present situation (before-after analysis) or between two different situations to understand the effectiveness of an intervention (with or without analysis). The strength of this model lies in the boundless capacity of ICT to store information in a retrievable manner and transmit it almost instantaneously across all geographical and hierarchical barriers.

7.5.4 Mobilisation and Lobbying Model

Mobilization and Lobbying Model is one of the most frequently used digital governance models and has often come to the aid of civil society organisations in developing countries to impact international decision-making processes. The model is based on planned, directed, strategic flow of information to build strong virtual allies to strengthen action in the real world. It takes up the proactive approach of forming virtual communities which share similar values and concerns, promoting active sharing of information between these communities and linking them with real-world activities. The strength of this model is in the diversity of its virtual community, and the ideas, expertise and resources accumulated through virtual forms of networking. The model is able to effectively overcome geographical, institutional and bureaucratic barriers to shape concerted action. It also provides a strong virtual arm to several activities such as directing campaigns against a particular individual or decision-making body.

7.5.5 Interactive-Service Model/Government-to-citizen-to-Government Model (G2C2G)

Interactive-Service Model in many ways is a consolidation of the earlier digital governance models and opens up avenues for direct participation of individuals in the governance processes. The model fully captures the potential of ICT and leverages it for greater participation, efficiency and transparency in the functioning of government as well as savings in time and costs relating to decision-making.

The Interactive-Service Model makes possible various services offered by the government to be directly accessible to citizens. It creates an interactive Government- to-Consumer-to-Government (G2C2G) channel in various functions such as election of government officials (e-ballots), filing of tax returns, procurement of government services, sharing of concerns and providing expertise, conducting opinion polls on public issues and grievance redressal.

7.5.6 E-Governance and Maturity Models

The E-governance Maturity Model (EMM-version 1.0) (D.C. Misra and Anjali Dhingra), based on the conventional software maturity models, proposes some levels of maturity, depending on the effectiveness with which the e-governance efforts have been initiated, implemented or successfully completed. The model also provides for identification of key focus areas that need to be concentrated for attaining a specific maturity level as discussed below:

Traditionally, e-governance has been defined as an ICT enabled route to good governance with a view to enhancing transparency in the system and provide prompt and quality services to the citizen. E-governance is an evolutionary path, whose effective implementation requires a complete understanding of its various elements and at the same time, taking a holistic view to stay focused on its overall objectives.

E-governance journey encounters several milestones that need to be identified and modelled so that efforts invested can be assessed and an appropriate course of action be taken by the organisation to further its way on the path of e- governance. The E-governance Maturity Model (EMM-version 1.0) proposes five levels of maturity, depending upon the effectiveness with which the e- governance efforts have been initiated, pursued, utilized and institutionalised. EMM 1.0 will facilitate government organisations to assess the current level of e-governance initiatives and accordingly make efforts for the future. The model further identifies the characteristics exhibited by organisations at various levels of maturity that will facilitate correct assessment of the current status. The model also provides Key Focus Areas (KFAs) that need to be focused on to attain a particular maturity level.

From the above discussion, it is evident that e-governance and e-government systems are intertwined and putting a clear boundary is quite complex. Besides, most of these models and frameworks do not elaborate participation of citizens at grassroots level nor elaborate a mechanism to capture them. The situation becomes more complex when we consider such initiatives for rural areas in the Indian context. Therefore, our emphasis on a prospective planning of *Village Information System* provides a direction towards providing better governance. In the prospects of IS planning for *Governance* and *Government* we consider that *government* needs to provide specialized services to the citizen through “superstructure of policies and programmes” while *governance* is defined as a superset of “government” through which citizen-centric policies, programmes are developed, implemented and monitored continuously. These policies and programmes are based on the citizen charters and receive inputs from the environment as well as through administrative structure of the government. It also monitors the performance of the government in broader terms. With this definition, purview of e-governance and e-government is more specific in the context of rural development.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the Unit

1. Lists any models of e-Governance in rural development?

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2. What is G2C2G? Explain its utilisations?

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3. Briefly explain the Gyandoot Internet Kiosks project?

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7.6 CASES IN RURAL E-GOVERNANCE IN INDIA

Some of the classic cases of rural e-Governance practices used in India are narrated below:

7.6.1 Case-1: Bhoomi: Online Delivery of land Title in Karnataka, India

The Department of Revenue in Karnataka has computerised 20 million records of land ownership of 6.7 million farmers in the state. Previously, farmers had to seek out the Village Accountant to get a copy of the Record of Rights, Tenancy and Crops (RTC) – a document needed for many tasks such as obtaining bank loans. There were delays and harassment. Bribes had to be paid with a nominal fee, a printed copy of the RTC can be obtained online at computerised land record kiosks. In the next phase, all the taluk databases are to be uploaded to a web-enabled central database. RTCs would then be available online at Internet kiosks, which are likely to be set up in rural areas.

7.6.2 Case-2: Gyandoot: Community – Owned Rural Internet Kiosks

In awarding the *Gyandoot* project the Stockholm Challenge IT Award 2000 in the Public Service and Democracy category the jury described it as “a unique government-to-citizen Intranet project ... with numerous benefits to the region, including a people-based self-reliant sustainable strategy. ‘Gyandoot’ is recognised as a breakthrough in e-government, demonstrating a paradigm shift which gives marginalised tribal citizens their first ever chance to access knowledge, with minimum investment.”.

7.6.3 Case-3: Gram Sampark

Gram Sampark is a flagship ICT product of the state of Madhya Pradesh. A complete database of the available resources, basic amenities, beneficiaries of government programmes and public grievances in all the 51,000 villages of Madhya Pradesh can be obtained by assessing the website *Gram Sampark*. It has three sections – *gram paridrashya* (village scenario), *samasya nivaran* (grievances redress) and *gram prahari* (village sentinel). An eleven-point monitoring system has been put in place whereby programmes are monitored village wise every month.

7.6.4 MS Swaminathan Foundation (MSSRF), Chennai

Since 1992, M.S. Swaminathan Research Foundation (MSSRF) has been implementing the Village Resource Centres (VRCs) and Village Knowledge Centres (VKCs). VRCs and VKCs mainly provide need-based locale-specific, demand driven information content (both dynamic and static) based on collection of several secondary data and a well-planned need assessment, organising training and awareness programmes and making linkages with several leading institutions/ organisations for translating the content into field-based applications. In 2003, the VRC and VKC programmes were further strengthened by the creation of the Jamsetji Tata National Virtual Academy (NVA) and Indian Space Research Organisation (ISRO)-VRC programme. This programme involves collaboration with several international and national partners for developing content and capacity to embark on sustainable rural development. This programme also leads to identifying and selecting grassroots people who involve rural transformation in the area of sustainable development and awarding the NVA Fellowship. During 2003, M S Swaminathan Research Foundation and many institutions launched a multistake-holder partnership network called “Mission 2007: Every Village a Knowledge Centre”. The purpose of the movement is to know how we will take ICT-enabled development activities to all over India particularly in the rural areas as a consortium mode. Since then every one of this network members will have annual meet and discuss many issues related to content, capacity building, care and management, connectivity, financial sustainability, etc. The mission is referred to as ‘Grameen Gyan Abhiyan – Rural Knowledge Movement’ since August 2007.

Under the CD-ROM library this programme provides necessary research inputs to researchers in the area of agriculture. The Hindu Media Resource

Centre of the MSSRF is organizing several theme-based media interactions, organising millennium lectures, etc. in different facets of sustainable development.

7.7 LET US SUM UP

Purpose of e-governance systems globally has remained same. But, implementations of such efforts have yielded varied results and success or failures of such efforts have remained limited to the nation. Generally, e-governance efforts have fallen short of expectations. National policies towards e-governance have not taken into consideration the advantages of ICT and other technologies fully. This scenario is largely attributed to the mismatch between the e-governance services being rendered and the requirements of the end users. In the context of rural development, the requirement of end users is quite dynamic and complex. In Indian environment, complexity has grown manifold because of socio-political systems, inadequate back end preparedness of government systems and above all, poor articulation of the issues and challenges faced by rural citizens. Thus, e-governance efforts need to focus on these issues in order to provide e-citizen services with holistic approach with the aim to address issues related to local livelihood systems, information asymmetry and market linkages.

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7.9 CHECK YOUR PROGRESS: POSSIBLE ANSWER

Check Your Progress: 1

1. 1. Improve service delivery mechanism;
2. Enhancing the efficiency in output;
3. Emphasising wider access to information;
4. Better management of accounting and record-keeping; and
5. Promote accountability and transparency

2.
 1. Economic dimensions
 2. Political dimensions
 3. Spatial dimensions
 4. Social dimensions
 5. Cultural dimensions
3. Due to the market demands, telecom infrastructure is much better than in rural areas. Equally, telecom users are much higher in urban than rural areas, leading to more digital infrastructure. Eventually, it created a spatial digital divide between urban and rural users.

Check Your Progress: 2

1.
 1. Broadcasting Dissemination Model
 2. Critical Flow Model
 3. Comparative Analysis Model
 4. Mobilisation and Lobbying Model
 5. Interactive Service Model
2. G2C2G stands for Government-to-Consumer-to-Government. This is part of the Interactive Service Model that makes various possible services directly accessible to citizens by the government. Its utilisation includes election of government officials (e-ballots), filing tax returns, procurement of government services, conducting opinion polls on public issues and grievance redressal system.
3. 'Gyandoot' is recognised as a breakthrough in e-government, demonstrating a paradigm shift that gives marginalised tribal citizens their first-ever chance to access knowledge with minimum investment. Gyandoot received the Stockholm Challenge IT Award in 2000.

UNIT 8 E-GOVERNANCE IN URBAN DEVELOPMENT

Structure

- 8.0 Introduction
- 8.1 Learning Outcomes
- 8.2 Need and Importance of e-Governance in Urban Development
 - 8.2.1 Need for E-Governance
 - 8.2.2 Importance of E-Governance
- 8.3 Initiatives of E-Governance: International Experiences
 - 8.3.1 Automated Building Plan Approval: Case of Singapore
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 - 8.3.3 An energy Efficient City: Madrid
 - 8.3.4 Urban Eco Map: San Francisco
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- 8.9 Check Your Progress: Possible Answers

8.0 INTRODUCTION

Generally speaking, e-governance or electronic governance in urban development is the use of information and communication technologies (ICT) in the operation and maintenance of urban services. ICT helps to introduce a wide range of ways in which government should use the Internet and computer networks for improving collaboration and cooperation between ministries; making government services more transparent, efficient and effective for the public by sharing accurate and up-to-date information and improving people's access to government services; boosting public sector accountability, transparency, efficiency and effectiveness. E-governance can also help streamline activities, cut costs and paperwork and help the city governments make more informed development decisions.

Many countries worldwide have been reforming their governments and public service delivery. They use ICTs as a key way to bring about urban

development. To illustrate, introduction of e-government initiatives is being perceived as a major tool in meeting the challenge that Africa is facing in bringing about city development. In many countries, for a simple transaction such as obtaining a form, a large number of citizens spend most of a day at a local administration office. Too often, they do not know fully about the range of available public services, what they are entitled to, or how the services are delivered. For civil servants, processing the transactions manually is a time-consuming exercise, which introduces human errors and corruption. Information that they collect is kept within particular departments and not shared among ministries and agencies. Decision makers do not have accurate and up-to-date information, analysis and forecasts on their country's socio-economic development, which are critical in making informed decisions based on evidence.

The initiative of the use of information technology (IT) in urban management started in the late nineties, especially after the adoption of the 74th Constitutional Amendment Act (CAA) in 1994, when urban local bodies (ULBs) became constitutional entities of local governance. Prior to this, local governance was the mandate of the state governments where the ULBs were supposed to perform certain functions mandated to them by the state governments. The passage of the 74th CAA resulted in the increased role of the ULBs in local governance. Further, the central government launched the programme of Jawaharlal Nehru National Urban Renewal Mission (JNNURM) in 2005 where adoption of reform in e- governance became mandatory for all the 65 Mission cities (all State capitals, all million plus (population) cities and cities of historical interest). The use of ICT is expected to help the ULBs and state governments to curb corruption, reduce time for the provision of civic services and bring about transparency in urban management. The 65 ULBs are in different stages of adoption of reforms and are committed to adopt a complete computerized system of service delivery by the end of the mission period.

8.1 LEARNING OUTCOMES

After studying this unit you should be able to:

- Discuss the meaning and concept of e-governance; and
- Describe various initiatives of e-Governance in different development sub- sectors in urban areas with the help of examples/case studies.

8.2 NEED AND IMPORTANCE OF E- GOVERNANCE IN URBAN DEVELOPMENT

8.2.1 Need for E-Governance

E-Governance which is a short form for electronic governance, also known as digital governance or online governance refers to the use of ICT to provide and improve government services, transactions and interactions with citizens, businesses, and other arms of government. While e-government is often thought of as *online government* or *Internet-based government*, many non-

Internet *electronic government* technologies can be used in this context, like telephone, fax, wireless networks and services etc.

E-Governance has become an essential tool for urban development by involving the use of IT in:

- Improving transparency;
- Providing information to the citizen speedily;
- Improving administrative efficiency;
- Improving public service such as transportation, power, health, water, security and municipal services.

One of the important components of e-Governance solution is the Geographical Information System (GIS). The GIS has helped in systematic mapping of four aspects:

- **Revenue mapping** – showing details of all municipal taxes and charges like property tax levied on various assets. This will help the officials by improving assessment and thereby increasing collection of revenues.
- **Municipal infrastructure mapping** – showing the details of municipal infrastructure like roads, solid waste management system, streetlight, housing, etc., thereby helping the city planners in town planning.
- **Resource and assets mapping** – showing specifically the assets for optimum utilization and hence providing better services to the citizens.
- **Poverty mapping**– showing socio-economic attributes of the poor households. This will help in targeting the correct citizens for the correct services.

8.2.2 Importance of E-Governance

Urban areas are currently the largest contributors to global energy consumption and climate change. The world's 20 largest cities alone – each with a population exceeding 10 million – are responsible for 75 percent of the planet's energy use. Added to this is the rapid development of metropolitan areas around the globe as well as the need to renew outdated 20th century infrastructures in cities. The scope of ICT in addressing these urban challenges is tremendous.

The further development of the information/knowledge/network society is now a common goal of many authorities round the world. Modern telecommunications can be seen not only as a new way of working but also as a new form of urban management. Delivery and management of urban services can be better done by e-governance. This may be true for property tax collection; vehicle tracking or efficient transport services. The quality and diversity of conventional transportation networks and services are important locational factors for many industries and activities. Therefore, analogously, one may suppose that ICT and its applications will, over the course of time, similarly come to affect spatial development and thus one may argue that

with this in mind, ICT should already now be taken into account in all future spatial planning processes. In the policies and plans for sustainability and eco responsibility in cities, much attention has been directed to three sectors: the built environment, energy, and mobility. At the commencement of the 21st century, it is obvious that a fourth, equally important element must be addressed: ICT.

This era is the start of a dialogue about how cities can create coherent, long term policies and plans to manage the environmental impacts of ICT and utilize ICT strategically to create sustainable 21st century cities. ICT products and systems are a significant and rapidly growing part of the environmental footprint of modern urban life. They are resource intensive in manufacturing and distribution, consuming ever greater amounts of energy while in use, and producing escalating volumes of solid and toxic waste.

ICT products, systems and networks are the essential drivers of productivity improvements and innovation for the 21st century. They will be the enablers of sustainability solutions in all networks of urban life: buildings, energy production and use, mobility, water and sewage, open spaces, education, and public health and safety. ICT innovation is also the catalyst for changes in personal, work and community life that will be a fundamental requirement for sustainable economic development.

Until now, little attention has been given to measuring the eco footprint of ICT in cities, or to clearly understand its role in enabling other sustainability initiatives. Because ICT systems and products are literally everywhere in modern life, it is difficult to see and understand that each device is part of a whole system globally linked by networks to create, manipulate, store, move and present information for humans and machines. To successfully manage the ICT environmental footprint and realize the benefits for enabling sustainability, a city must have a vision and strategy for ICT that encompasses all organisations and constituencies.

ICT helps develop an innovative combination of advanced technologies that meet the unique needs and vision for each urban area. Some of these technologies include:

- Vehicle tracking/identification systems;
- Dynamic congestion-charging programmes;
- Video communication solutions;
- Integrated transportation management systems;
- Global positioning system (GPS), radio frequency identification (RFID) and other sensor technologies;
- Broadband, wireless and intelligent infrastructures;
- Collaboration technologies in the creation of innovative work environments;

A study about the relationship between the quality of local e-government services and the levels of Internet access in the 12 regions of the United Kingdom reveals that lower quality of local e-government services correlate

with low levels of Internet access. In Taiwan, electronic tax-filing systems, the implications of technology acceptance and perceived credibility of the systems are positive factors that influence adoption of e-government services. Whether e-government in the future will be a method for including more citizens in a government or excluding less technologically educated citizens remains a concern. Many information policy issues are likely to present significant challenges to the development of e-government. These are:

- Ensuring ability to use required technologies;
- Educating citizens about the value of e-government;
- Ensuring access to useful information and services;
- Coordinating local, regional and national e-government initiatives;
- Developing methods and performance indicators to assess the services and standards of e-government;
- Providing consistent and reliable electricity, telecommunications, and Internet access;
- Addressing issues of language and communication;
- Preventing e-government from lessening responsiveness of government officials;
- including individuals with disabilities in e-government

8.3 INITIATIVES OF E-GOVERNANCE: INTERNATIONAL EXPERIENCES

Examples of e-Governance practices followed in some of the developed countries are given below:

8.3.1 Automated Building Plan Approval: Case of Singapore

Singapore is a key hub in the development of the global information and knowledge economy. Part of the developing e-Business capability in Singapore is the Construction and Real Estate Network project (CORENET). This world's leading initiative has changed the entire perception of building planning in Singapore. Key to the success of CORENET is Automated Code Checking, which allows automated approval of building plans over the Internet. Received data is stored and checked within EXPRESS Data Manager software.

CORENET is a major IT initiative undertaken by the Singapore Ministry of National Development to re-engineer the business processes of the construction industry and achieve a quantum leap in turnaround time, productivity and quality. CORENET centres on developing IT systems to integrate the four major processes of a building project life cycle, supported by key infrastructures, to provide a *One-Stop Submission Centre* (OSSC) to facilitate electronic submission, processing and approval of building project documents over the Internet.

The specific objectives of CORENET are:

- To speed up the business planning and project evaluation processes in the building industry by making all relevant information readily available to all players in the industry.
- To streamline the design-related processes by having all industry professionals work with one common set of electronic plans instead of separate paper and electronic plans presently used.
- To evaluate the compliance of building design to statutory requirements with minimal intervention and guidance from the regulatory authorities.
- To move towards common procurement procedures and document standards.
- Integrated Building Plan and Building Services System (IBP/IBS).

One of the major projects in CORENET is the IBP/IBS which is an expert system that automatically performs checks on digital plans for compliance with building plan and building services regulatory requirements. The design checking and approval process using the manual approach is time-consuming and inefficient. Automating this process eliminates potential delays as well as avoids inconsistencies in code interpretation.

One-Stop Plan Submission of Building Plans

It is a virtual submission centre to facilitate round the clock electronic submission and approval of digital building plans. It will be an advanced e-government solution for industry professionals such as architects, structural, mechanical and electrical engineers. Automated compliance checks are performed on the relevant parameters of the proposed building against the building codes and compliance regulations. Payment of fees is electronic. The system saves the industry travelling time to the 13 regulatory agencies and brings about significant manpower and cost savings to both industry and agencies.

8.3.2 Smart Work Centers: Amsterdam

This network of Smart Work Centers (SWC) is part of collaboration between Cisco and the City of Amsterdam under the Connected Urban Development program (CUD), which serves to address modern urban challenges including mobility, climate, energy and sustainable ways of urban management. The Double U Smartwork aims at a one stop shop for location providers, individual users, independent professionals and corporate users, where all SWCs can be booked directly by a central booking tool.

SWCs comprise a regional network of neighbourhood professional work and community centres supporting travel virtualisation and enabling mobile working practices. SWCs offer a professional work environment near residential areas to lower energy use and carbon emissions. Estimates revealed that users have saved on an average of 66 minutes of commuting time per day.

An SWC is a physical facility where high quality workplace solutions are offered to professional workers in a neutral, centrally located and easily accessible environment. To minimize traffic, an SWC is located in the vicinity of roads, traffic junctions, stations and residential areas. Currently, there are several providers of SWC-like facilities in the Netherlands. The Double U Smartwork Foundation serves as a coordinating platform for SWC providers and aims to develop a national network. Users, regardless of where they live or reside, should have access to a good workplace within biking distance. Until recently, existing SWCs were too fragmented and locally focused, therefore, employers operating on national level were not interested in offering their employees an alternative working spot. Double U links a network of over 50 open and SWCs, with the plan to extend to 100 national centres.

Smart Work Centers are well equipped and go beyond providing a workplace. The services and facilities are not only meant to facilitate work itself, but also to provide work related services as day-care and catering facilities.

Anyone can use the facilities offered by one simple online booking system. Through the portal, users can quickly find the nearest location with the right facilities, whether they are on the road or at their workplace. The available providers are automatically displayed.

8.3.3 An Energy Efficient City: Madrid

Madrid is one of the first pilot projects carried out in Spain within the Connected Urban Development program (CUD), in which companies and cities partner to contribute to the development of sustainable, efficient and innovative cities through the use of connectivity and new technologies. Promoted by the Municipal Company for Housing and Lands of Madrid; Cisco and technology partner Telvent, have deployed network infrastructure, connectivity and control systems in a pilot, apartment building in the city. The development is intended as temporary housing on a rental basis to young people in Madrid.

The *Energy Efficiency Manager* installed in homes can, at any time and in real time, manage energy consumption, controlling emissions of carbon dioxide and make decisions about the way in which residents make use of energy both at the individual apartment level and throughout the building. In the future, this is intended to extend across the urban community. The solution, which allows consumers to set limits and comparisons of weekly, monthly or yearly consumption, provides to citizens and municipal managers, daily tips to improve efficiency and be more environmentally responsible.

Urban Eco-Map is part of the global Urban Services Platform approach toward which visionary cities and the ICT industry are moving. Urban Eco-Map provides real-time environmental intelligence to enable citizens, communities, cities, countries and businesses alike to make smart ecological decisions and to develop policies that improve the sustainability of cities. Through this comprehensive view of eco-data, we can now take a global *pulse* of the eco-health of our planet.

8.3.4 Urban Eco Map: San Francisco

Urban Eco-Map: A pilot co-developed with the City and County of San Francisco, Urban Eco-Map provides cities with relevant data regarding primary greenhouse gas contributors – transportation, waste and energy – to help city residents take action to reduce their emissions.

8.3.5 Personal Travel Assistant: Seoul

Personal Travel Assistant (PTA): PTA is a Web-based service that allows residents in Seoul and Amsterdam to make on-the-go travel decisions based on time, cost and carbon impact. It offers *virtual assistant* features that provide transit guidance based on user preferences via any Web-enabled device, from any location.

Check Your Progress: 1

- Note:** 1) Use the space below for your answers.
2) Compare your answers with those given at the end of this Unit.

- 1) What do you mean by e-governance?

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- 2) Explain the importance of e-governance and how e- governance has helped urban management across countries with examples.

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8.4 INITIATIVES OF E-GOVERNANCE: NATIONAL EXPERIENCES

Examples of E-Governance practices in some of the Indian states are narrated below:

8.4.1 Computer-Aided Registration of Deeds and Stamp Duties: An initiative of the Andhra Pradesh Government

E-governance has helped the state revenue departments to register properties in less time, store information scientifically and bring about transparency in

the system. Such an initiative was taken in Andhra Pradesh in the late eighties, where the Computer-aided Administration of Registration Department (CARD) project has brought about computerised counters at land registration offices throughout Andhra Pradesh. The project aimed at altering the antiquated procedures that had governed the registration system of the state, which included the laborious copying and indexing of documents as well as their unscientific space-consuming preservation in ill-maintained backrooms. The state had a flourishing business of brokers and middlemen who exploited citizens selling or buying property. The CARD project is an attempt to reform this system through the use of IT¹. With the introduction of CARD, citizens now complete registration formalities within a few hours. The CARD project illustrates some of the key implementation issues the state and national governments may face in their efforts to use IT to improve citizen-government interfaces and serves as a best practice to be replicated by them (J. Satyanarayana, 2002).

The idea of introducing computers originated in 1988 in Andhra Pradesh when a project was initiated to computerise the process of issuing Encumbrance Certificates. A 386 server operating with 14 terminals was set up at a cost of about \$31,000 (Rs 1.33 million). Data entry of index registers of the twin cities of Hyderabad and Secunderabad was initiated. The National Informatics Centre (NIC) provided the technical assistance. The entry of 15 years of data went on until 1995 when a pilot scheme for issuing computerized Encumbrance Certificates was launched in one of the city offices. The feasibility of taking up a comprehensive Registration Department computerization project to address other registration formalities and problems was established in a study conducted by J. Satyanarayana in August 1996. The study brought out methods by which the various registration services could be delivered electronically across the counter in an integrated manner and showed a road map as to how the process of valuation could be consigned to the computer and also introduced the concept of electronic document management as an essential part of computerising the registration process.

Objectives of the CARD Project

CARD is a major IT project designed to eliminate the maladies affecting the system of registration through electronic delivery of all the registration services. It was based on the primary objectives outlined below.

1. The CARD project was funded entirely by the government of Andhra Pradesh. The original outlay was about US\$3 million.
2. Commissioner & Inspector General of Registration and Stamps C.T. & Excise Complex, M.J. Road, Nampally, Hyderabad
 - Demystify the registration process;
 - Introduce a transparent system of valuation of properties, easily accessible to citizens;
 - Bring in speed, efficiency, consistency and reliability;

- Replace the manual system of copying and filing of documents with a sophisticated document management system that uses imaging technology;
- Replace the manual system of indexing, accounting and reporting;
- Introduce electronic document writing; and
- Substantially improved the citizen interface.

Benefits of CARD

The CARD project aims at providing improved quality of the services at the registration department by providing a computer interface between citizens and government. The tedious procedures that took weeks have been replaced by a system that can be accomplished in just a few minutes. The market value assistance and issuing of the Encumbrance Certificate (EC) takes five minutes each. The sale of stamp papers, document writing and registration of the documents takes ten minutes, thirty minutes and one hour respectively. The positive impact of the CARD project on the efficiency of registration operations can be gauged from the following table:

8.4.3 Kaveri in Karnataka

KAVERI is another e-governance initiative undertaken by the Government of Karnataka. For the last five decades, the process of registration of documents was done manually and involved the following steps:

- Stamping,
- Presentation,
- Admission of execution,
- Identification by witnesses and
- Registration, as prescribed in Karnataka Stamp Act, 1957 and Registration Act, 1908

Features of KAVERI

Automated Kiosks with touch screen operation facility were installed in every Sub-Registrar's Office, through which public can have access to the following information in Kannada and English.

- Market value of land in all villages, towns and cities in the state.
- Model formats of commonly used deeds and forms required for Registration of Marriage.
- Model byelaws of Societies and Associations.
- Frequently asked questions and exhaustive answers.
- Fee for Registration of documents/Registration of Societies/Firms/Marriages.
- Acts and Rules bearing on registration of documents.

Under the manual registration process, the documents registered were copied manually in specified books. After that, they were verified with the original documents, and the hand written documents were authenticated by

Registering Officers. The registered book would serve as a public document. Since the manual procedure involved writing each document that was to be registered, the time taken for the entire registration procedure was anywhere between two to three months. Moreover, it also meant 2–3 trips to the registration office to check if the document was ready. The solution lay in finding an alternative procedure that would meet the statutory requirements and also speed up the process while preserving the accuracy of the manual procedure.

Computerisation was the way forward. The Department of Stamps & Registration, Government of Karnataka set up automated registration process in the state in the year 2002. More than 200 Sub-Registrar Offices in Karnataka came under computerisation under an outsourced model whereby the vendor could complete the registration process within 30 minutes. The software was aptly called KAVERI, after the river Cavery.

After the introduction of KAVERI the department has registered documents and returned the same to the parties concerned within 30 minutes of its presentation. The graph indicates that there was a significant growth in the revenue to the state exchequer after the introduction of KAVERI. In spite of reduction in stamp duty and registration fee there was significant growth in revenue. This was possible due to the elimination of the human interface as the KAVERI system calculates the value of property through automated programme.

8.4.3 E-Suvidha-E-Governance Initiatives of Pimpri Chinchwad Municipal Corporation

The Pimpri Chinchwad Municipal Corporation (PCMC) has introduced an integrated e-Governance Programme to ensure improved transparency to build citizen centric governance. The implementation of e-governance gave new set of responsibilities to PCMC's employees. This project has involved around 11 corporation departments, which are computerized. Citizen Facilitation Centre (CFC) is the most successful project, which provides services to citizens with more than 99% efficiency. Another Innovative project is PCMC@home, which provides services of corporation directly at home through PCMC personnel.

The initiative under E-Suvidha has considerably reduced the hassles faced by the citizens. Citizens take minimum time and cost for availing the civic services. Citizens avail the facility on a mouse click from home or office or any remote location-avoiding visit to Corporation office or division office. Payment of charges and taxes directly online through payment gateway facilities reduces the travelling cost and valuable time. Citizens save about 3–4 hours of time through availing E-Suvidha facilities and in some cases citizens save days by availing the online facilities of the PCMC.

Kiosks set up in different wards, zonal offices and other public places have helped the citizens who are not proficient with net banking and who are not able to avail web based facilities. Visit to Kiosks at the nearest point helps the citizens to avail the facilities, services and payment of different charges,

taxes and bills. The web based application helps the citizens to locate their properties for assessment details and for payment of taxes online. The utility mapping has helped the PCMC to monitor the delivery of essential services such as water supply, drainage lines, roads, streetlights, garbage bins, etc. This has increased the overall service delivery improvement to provide services and identify the areas which do not avail these services and utilities.

1. Centre for Development of Advanced Computing, Pune (C-DAC) provided the technical support in developing suitable software to cover the following aspects of registration: Registration of properties, Valuation of properties, Scanning and Archival of Documents, Reports, Vendor management system, Utilities, Website, Societies, Firms and Marriage Registration and Data Transmission.
2. The Government of Karnataka reduced the rate of stamp duty on conveyance from 10% to 8% and Registration fee from 2% to 1% with effects from 01-04-2003.
3. This initiative of the department got two e-governance awards from GOI for introducing computerization as best practice.

The following section provides details of the services provided under E-Suvidha initiative:

- **Property and water revenue management:** The property assessment details and water charges are available online, with online payment facility. This has reduced number of visits by the citizens to the corporation office.
- **e-Tendering:** PCMC has initiated the online tendering system for all projects and procurements to be taken up for the development works under its limits. Submission of tenders and documents can be done online.
- **Dashboard for works management:** Dashboard of work management is an integrated web based software for monitoring and tracking the progress of work. This module is also integrated with financial data like budget approved for works, cost incurred and other information, which helps in keeping a track of the projects undertaken by PCMC. Necessary decisions are taken from time to time depending upon the progress of projects and works.
- **Citizens Facilitation Centre (CFC):** PCMC's Citizens Facilitation Centre (CFC) provides 79 different citizen centric services for 12 departments of the Corporation. The CFCs work on single window basis to provide one stop service to the citizens for PCMC. CFCs also provide services of the District Collectorate like caste certificates, domicile certificate, ration card, and also provides value added services of Road Transport Office like issuance of learning licenses and collection of Maharashtra State Electricity Board bills, collection of BSNL bills, collection of insurance premiums and railway ticket booking.
- **SMS-based complaint monitoring system:** To reduce the difficulties of citizens and to send a complaint to PCMC, a SMS based complaint

system has been initiated. A citizen can send a complaint by SMS and scrutiny of received complaints takes place through PCMC administration. An SMS as well as an email immediately goes to the related officer for addressing the complaint. This has reduced PCMC's response time considerably.

- **Solid waste management with vehicle tracking** PCMC has also started GPS vehicle tracking system. This GPS system has been integrated with an interface, which will assign waste pick up job and duty management. The system also monitors and registers the auto job picks adherence via geo reference and stop at pick up bin location. Vehicles trip/job report gets generated for number of trips per vehicle per driver and as well as contractor. Pick up adherence report; exception report on missed bins also gets generated for the authority to monitor the collection of solid waste from bins. Tracking report, stoppage, over-speed reports, detention reports etc., are getting generated for continuous monitoring of collection and transportation of vehicles.
- **Geographical Information System:** PCMC has a GIS mapping of 182 sq. km area. This has been done through geo-referencing of the Quick Bird satellite map of 0.6 m resolution map. This mapping has been developed for GISDA by Science and Technology Park (STP) and on terms of integration with various databases and application services. GISDA runs from a centrally located system, which can be accessed through web. GISDA provides core web technology and a GIS platform that is used by all other applications to provide Web-GIS based Citizen Centric Services.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) Taking Pimpri Chinchwad Municipal Corporation as a case study, name various initiatives of e-Governance taken in different development sub-sectors in the city.

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8.5 CHALLENGES IN E-GOVERNANCE

Though this unit has presented discussions of interesting and innovative e-government initiatives, e-government still faces many challenges as it continues to develop. In designing and implementing e-government sites, a government must consider elements of policy, including regulatory issues, economic issues, and the rights of users.

One US General Accounting Office report specifically listed the challenges

to implementing e-government as (1) sustaining committed executive leadership, (2) building effective E-Government business cases, (3) maintaining a citizen focus, (4) protecting personal privacy, (5) implementing appropriate security controls, (6) maintaining electronic records, (7) maintaining a robust technical infrastructure, (8) addressing IT human capital concerns, and (9) ensuring uniform service to the public. Other scholars have noted additional broad challenges, such as defining the parameters of e-government and making e-government function so that it does not conflict with other laws.

A recent U.S. government study found that the biggest concerns for e-government managers were not technical issues, but instead were policy issues, including coordination and collaboration between agency leaders, agency-centric thinking rather than focusing on the overall goals and functions of e-government, and communication to better understand and foster inter-relationships between e- government projects. Along with complex policy issues, some of the greatest challenges to maximising the potential of e-government may involve social dimensions of information policy related to the Internet. The planning and implementation of e-government, as it continues to develop and grow around the world, will have to focus on finding methods to address varied issues. Some of the most important sources of information about meeting challenges to effective e-government are actual e-government initiatives that are currently operational. The lessons that can be learned from ongoing e-government projects, both in what works and what does not, will provide meaningful guidance in developing and refining e-government. Furthermore, the examination of e-government projects from different levels of government and different parts of the world offers a method to share knowledge about e-government. In many ways, the future directions of e-government will be confronting the important policy issues that remain unaddressed. Studies such as those in this symposium issue are valuable to the conceptualization and application of current and future e- government projects, regardless of where the projects occur.

The Government of India has launched the National e-Governance Plan (NeGP) with the intent to support the growth of e-governance within the country. E- government helps simplify processes and makes access to government information easy citizens as well as for public sector agencies.

8.6 LET US SUM UP

E- governance or electronic governance in urban development is the use of information and communication technologies (ICT) in the operation and maintenance of urban services. This initiative of the use of information technology (IT) in urban management started in the late nineties, especially after the adoption of the 74th Constitutional Amendment Act (CAA) in 1994, when urban local bodies (ULBs) became constitutional entities of local governance. Further, the central government launched the programme of Jawaharlal Nehru National Urban Renewal Mission (JNNURM) in 2005 where adoption of reform in e-governance became mandatory for all the 65

Mission cities (all State capitals, all million plus cities and cities of historical interest). The Government of India has launched the National e-Governance Plan (NeGP) with the intent to support the growth of e-governance within the country. The introduction of e-governance has facilitated the state and local governments in the country in successfully delivering urban services to its citizens in a transparent and efficient manner. Although e- governance has been introduced in all the Mission Cities under JNNURM, concerted efforts need to be taken to increase the coverage across and within cities. Efforts should also be taken to include non-Mission cities under e- governance.

8.7 KEY WORDS

E-governance: E- governance or electronic governance in urban development is the use of information and communication technologies (ICT) in the operation and maintenance of urban services.

JNNURM: Launched in 2005, Jawaharlal Nehru National Urban Renewal Mission (JNNURM) is a central government reform linked programme wherein adoption of reform in e-governance is mandatory for all the 65 Mission cities (all State capitals, all million plus cities and cities of historical interest).

CARD: The CARD is a project aimed at altering the antiquated procedures that had governed the registration system of the state of Andhra Pradesh, which included the laborious copying, and indexing of documents as well as their unscientific space-consuming preservation in ill-maintained backrooms. The state had a flourishing business of brokers and middlemen who exploited citizens selling or buying property. The CARD project is an attempt to reform this system through the use of IT.

KAVERI: The Department of Stamps & Registration, Government of Karnataka set up automated registration process in the state in the year 2002, wherein 202 Sub-Registrar Offices in Karnataka came under computerization whereby the vendor could complete the registration process within 30 minutes. The software was aptly called KAVERI, after the river Cavery.

E-Stamping: To prevent fraudulent practices in stamp paper based transactions and registrations, e-stamping has been introduced in some states where the entire exercise of stamp duty payment and generation of stamp duty certificate takes less than 3 minutes. It offers a secure and reliable stamp duty collection mechanism, and stores information in secured electronic form and builds up a central data repository to facilitate easy verification.

E-Suvidha: Development of an integrated e-Governance programme to ensure improved, transparent and efficient way of building citizen centric governance.

8.8 REFERENCES AND FURTHER READINGS

JNNURM programme, Ministry of UD, GOI, www.urbanindia.nic.in
Municipal e-Design Document, Ministry of UD, GOI, www.

8.9 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) E- governance or electronic governance in urban development is the use of information and communication technologies (ICT) in the operation and maintenance of urban services.
- 2) Two examples of the use of IT in the field of property registration are CARD in Andhra Pradesh and KAVERI in Karnataka. The CARD is a project aimed at altering the antiquated procedures that had governed the registration system of the state of Andhra Pradesh, which included the laborious copying, and indexing of documents as well as their unscientific space-consuming preservation in ill-maintained backrooms.

Check Your Progress 2

- 1) Property and water revenue management; e-Tendering; Dashboard for works management; Citizens Facilitation Centre (CFC); SMS-based complaint monitoring system; Solid waste management with vehicle tracking; Geographical Information System; Property and water revenue management; e-Tendering; Building permission management.

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